

Lecture 4: Physiological Sensor Data

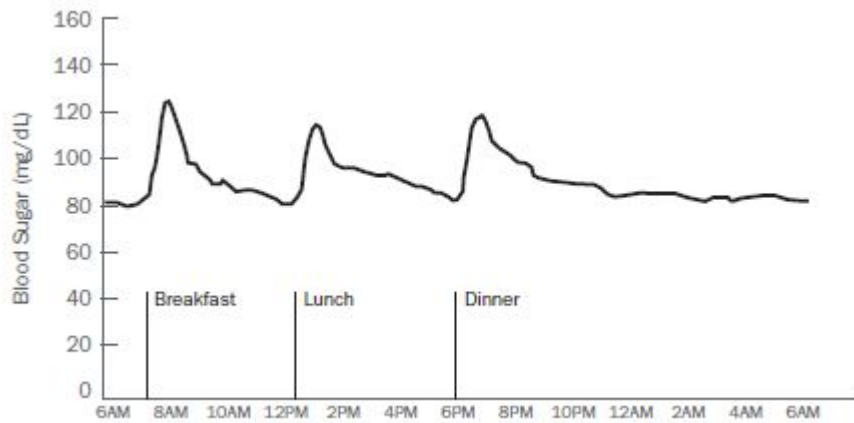
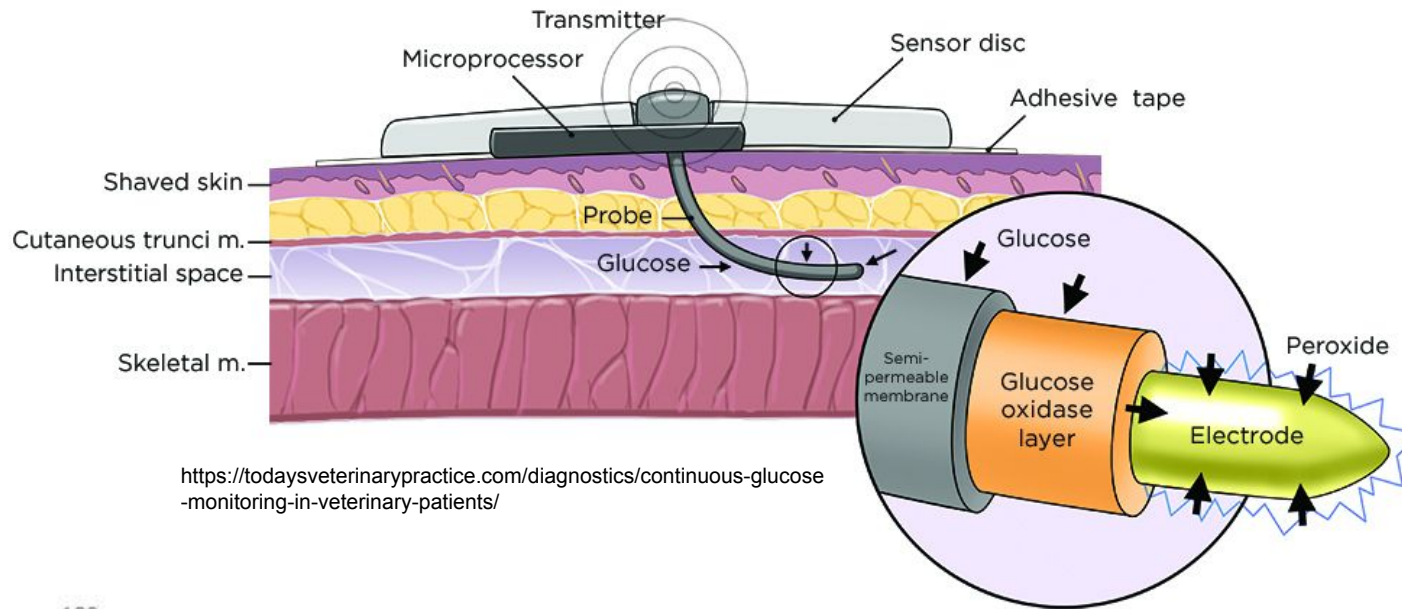
CSCI6410/EPAH6410/CSCI4148

Finlay Maguire (finlay.maguire@dal.ca)

Learning Overview

- Types of medical sensor data
- Attributes of Signals
- Challenges of Signals
- Preprocessing
- Multiple sensors
- Time-domain approaches
- Frequency & time-frequency
- State-space approaches
- Self-supervised learning and foundation models
- Examples:
 - Sound Cough/COVID detection
 - ECG beat segmentation
 - EEG seizure prediction

Physiological sensors (typically) capture data over time

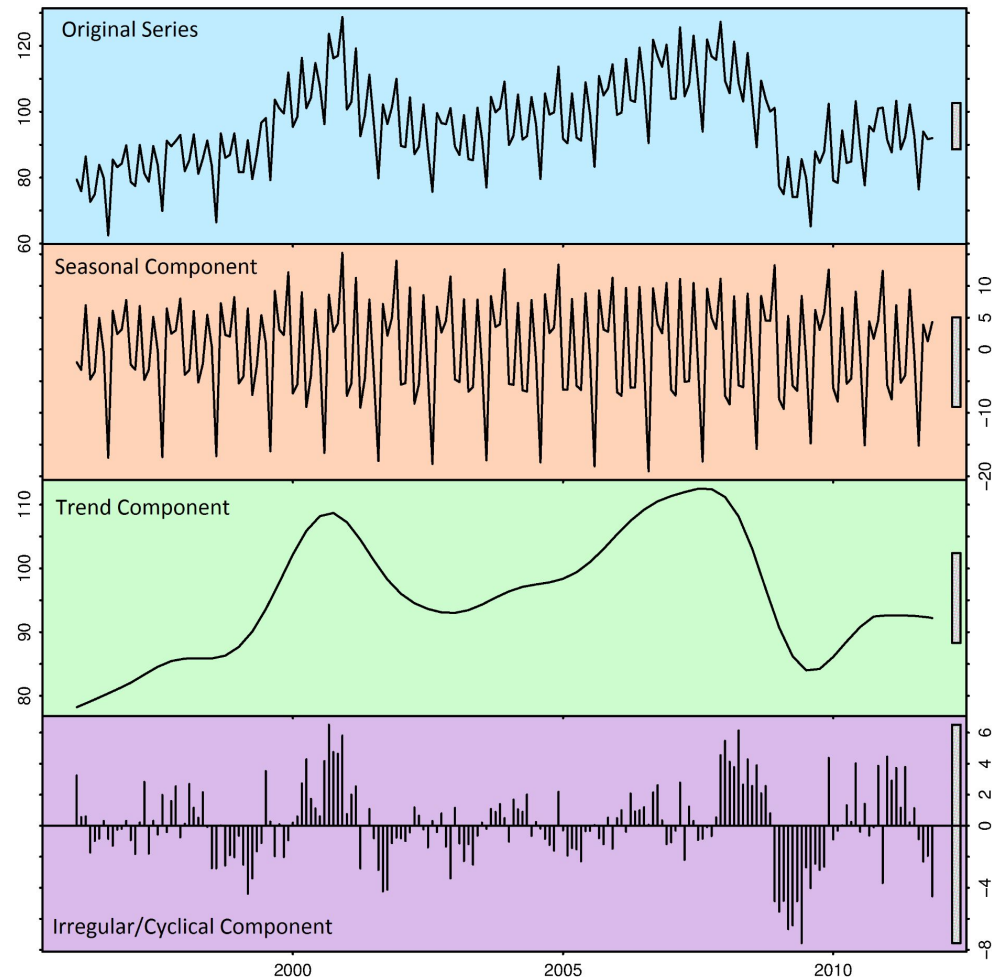


<https://www.diabetesdaily.com/learn-about-diabetes/understanding-blood-sugars/is-my-blood-sugar-normal/>

- Time-series: set of values ordered by time
- Single object observed over time vs cross-section of multiple objects on common time axis
- Simple 1-dimensional variable: continuous glucose monitoring

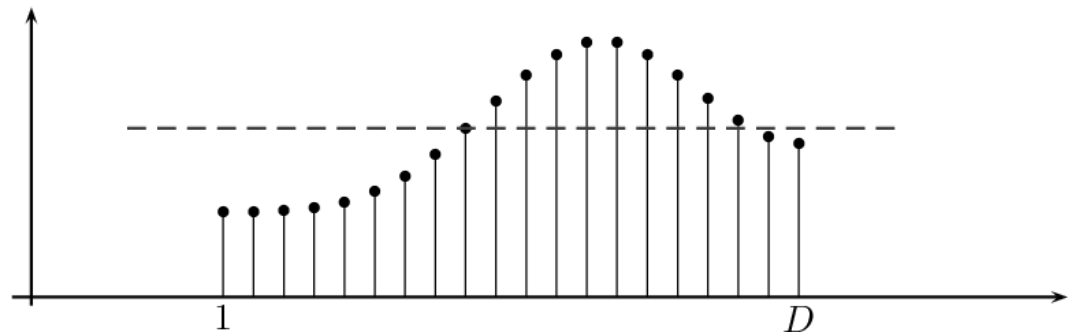
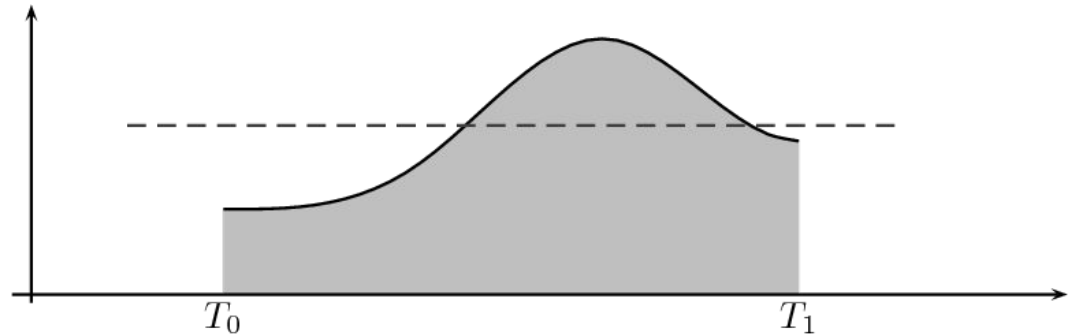
Time-series often have multiple components

- Short period fluctuations: seasonality
- Long period fluctuations: cycles
- Long-term directionality: trend
- Noise: stochasticity
- Correlation in successive times: autocorrelation
- Distribution changes over time: stationarity/non-stationarity



Time can be a discrete or continuous value

- Continuous time: defined at every real value of T
- Discrete time: defined at discrete intervals of T
- Real-world data is continuous but sampled during collection as discrete data (sampling rate)
- Conversion to some fidelity is possible
- Impacts analysis methods

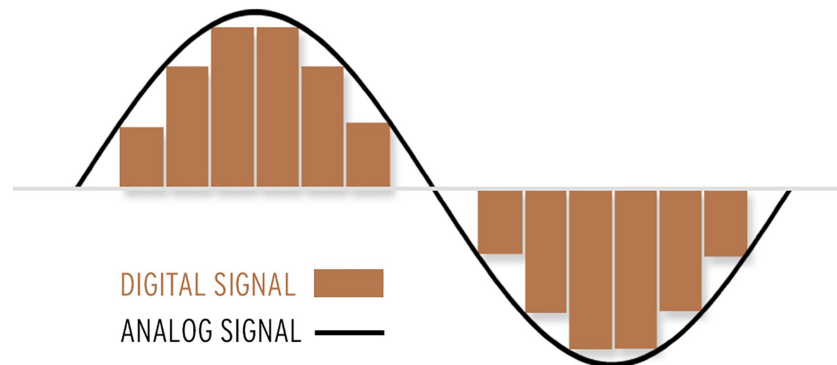


Physiological sensors capture signal data

- “Signal” is a broadly defined term
- For today: signals are analogue or digital representations of analogue physical quantities.
- Typically electrical representations created by a transducer
- Data encoded in voltage, current and/or frequency
- Medicine: often directly capturing bioelectrical signals



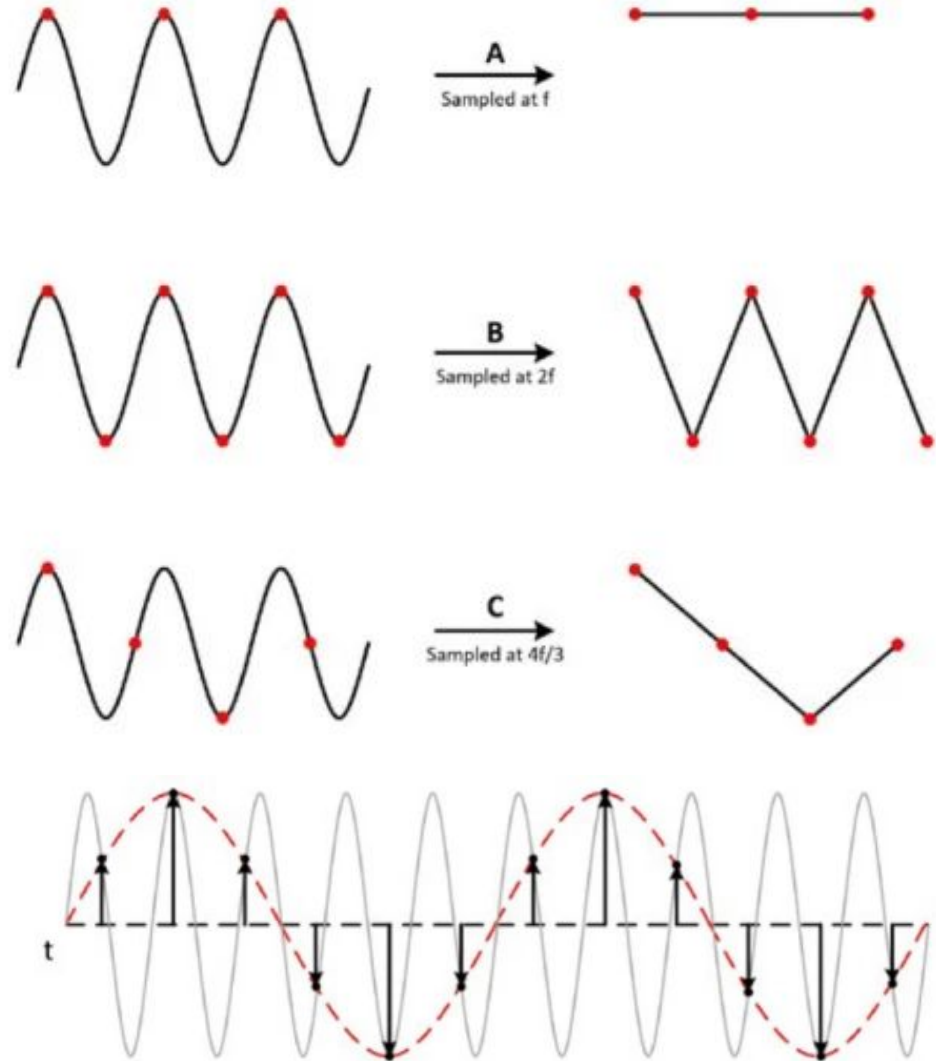
<https://mynewmicrophone.com/how-do-microphones-work-a-helpful-illustrated-guide/>



<https://www.klipsch.ca/blog/digital-vs-analog-audio>

Sampling rate bounds capture: Nyquist-Shannon

- **Nyquist-Shannon Theorem:** Must sample $>2\times$ the highest frequency to reconstruct a signal
- **Nyquist frequency** = half the sampling rate i.e., maximum capturable frequency
- **Aliasing:** under-sampled high frequencies indistinguishable from low frequencies
- **Anti-aliasing:** filter out high frequencies as part of signal capture and sampling



So, what are the most common types of physiological sensors?

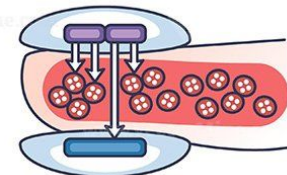
Wearable consumer sensors: smart watches

- **Photoplethysmography:** optical sensor measuring blood-volume change
- Infer heart rate, heart rate variation, some arrhythmias, & respiration rate

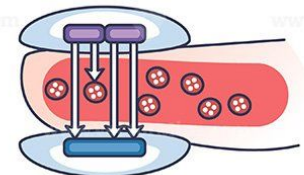


- **Pulse oximetry** (SpO_2): ratio of red/infrared absorption.
- Infer blood oxygen saturation

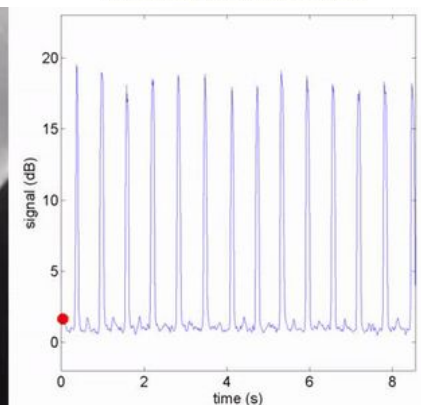
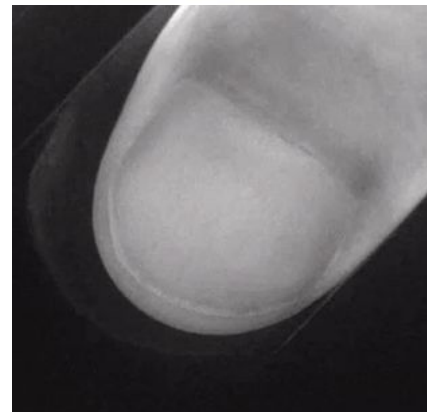
✓ HIGH OXYGEN - SpO_2 95-100%



✗ LOW OXYGEN - SpO_2 < 95%

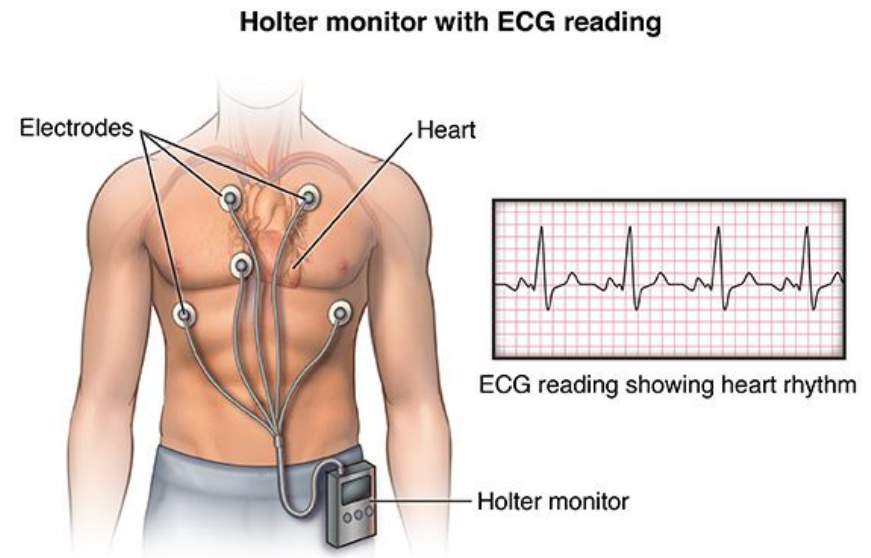


- **Accelerometry:** 3-axis motion capture
- infer gait, sleep/wake, fall detection, step counts
- Relatively cheap, common, and long timescale of capture
- **But** noisier and less calibrated than clinical sensors!



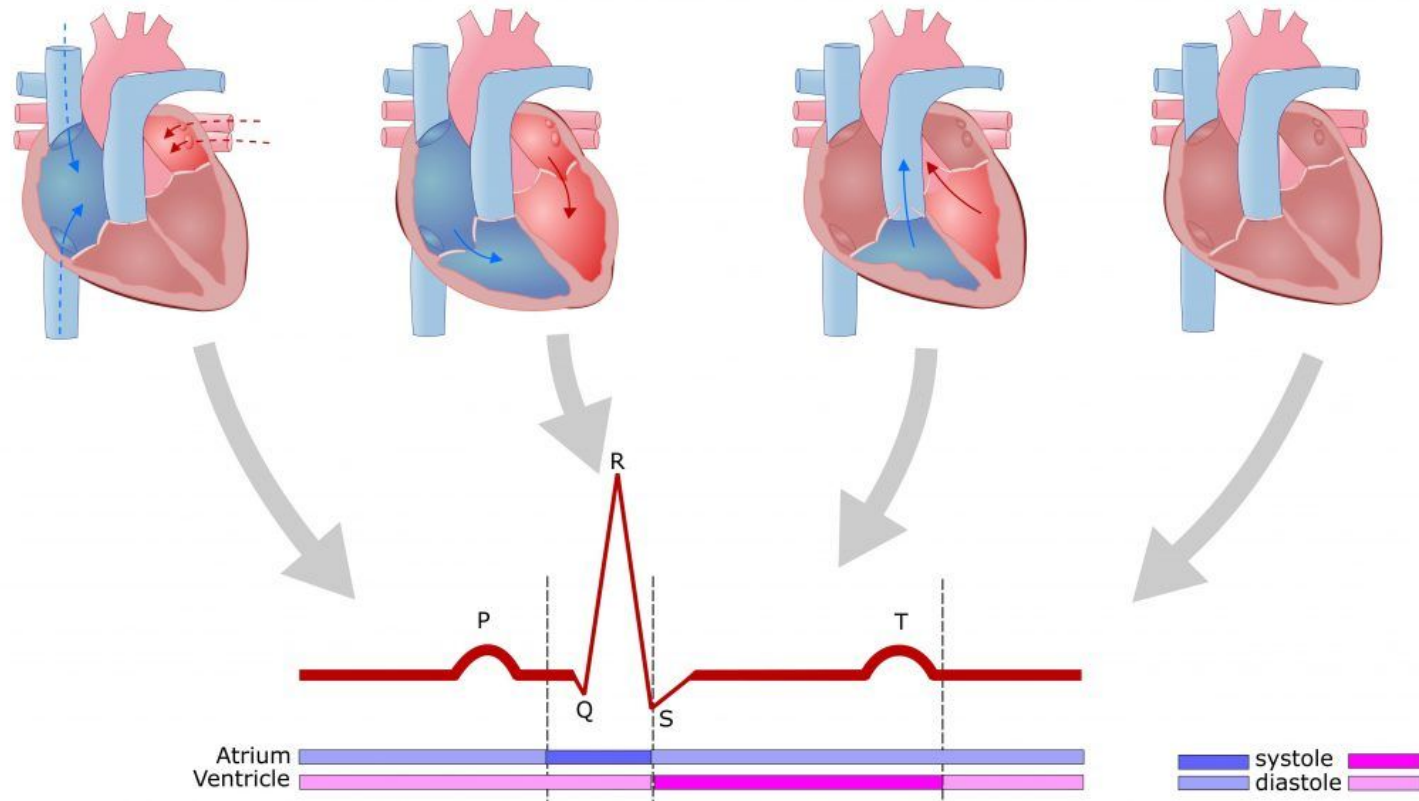
Electrocardiogram (ECG): cardiac electrical signals

- Recording of heart's electrical activity using multiple electrodes
- Signal from cardiac muscle {de,}polarisation during systole (contraction) and diastole (relaxation).
- Changes in pattern indicate abnormalities (e.g., rhythm disturbances, coronary blood flow, electrolyte disturbances)
- **Heart Rate**: number of cycles within period (bpm)
- **Inter-Beat Interval**: time between cycles (ms)
- **Heart Rate Variability**: variation in IBI



<https://www.hopkinsmedicine.org/health/treatment-tests-and-therapies/holter-monitor>

Electrocardiogram (ECG): inscrutable defined components



P-wave: depolarisation of atria -> atrial systole

PR-interval

QRS complex: atrial diastole -> depolarisation of ventricles -> ventricular systole

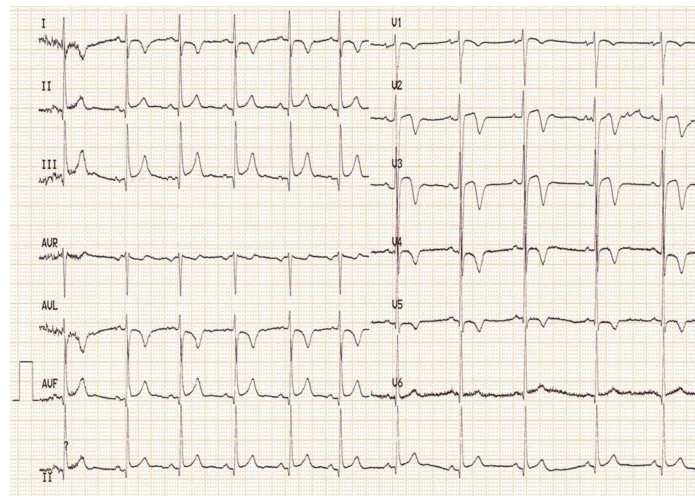
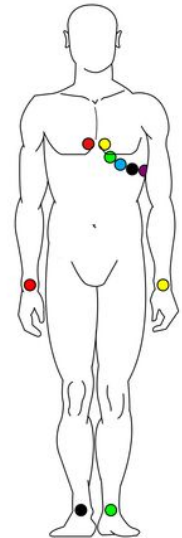
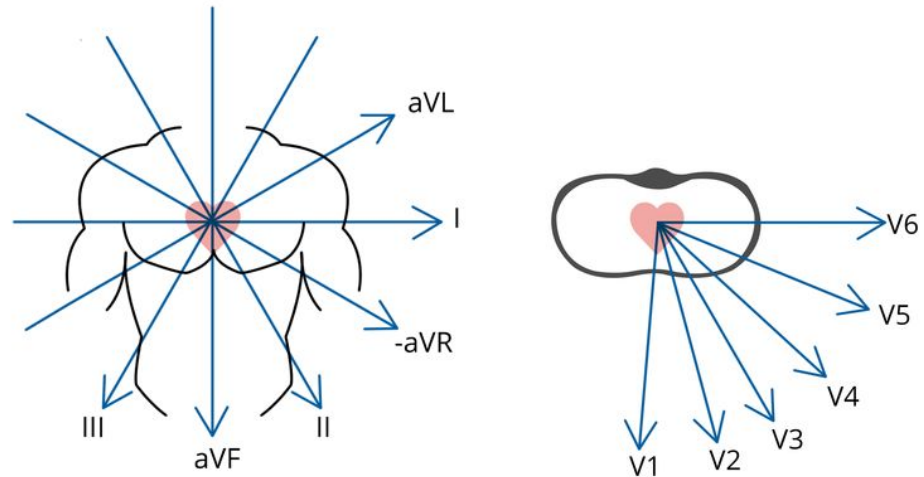
ST segment

T-wave: repolarization of ventricles -> ventricular diastole

TP segment

Multiple sensors recording same signal improves data

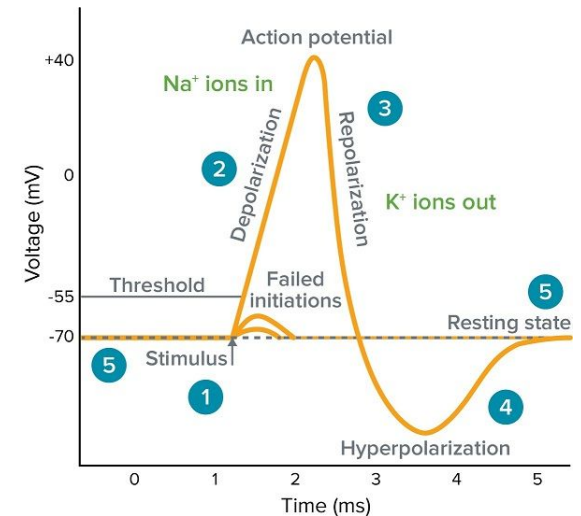
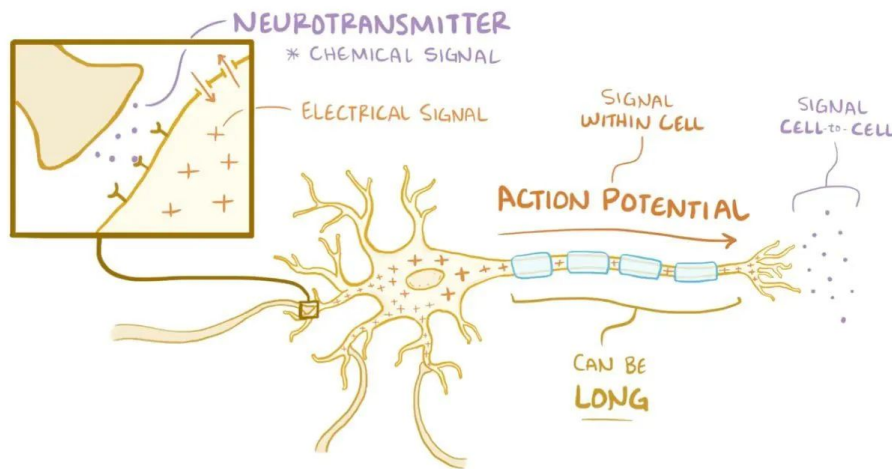
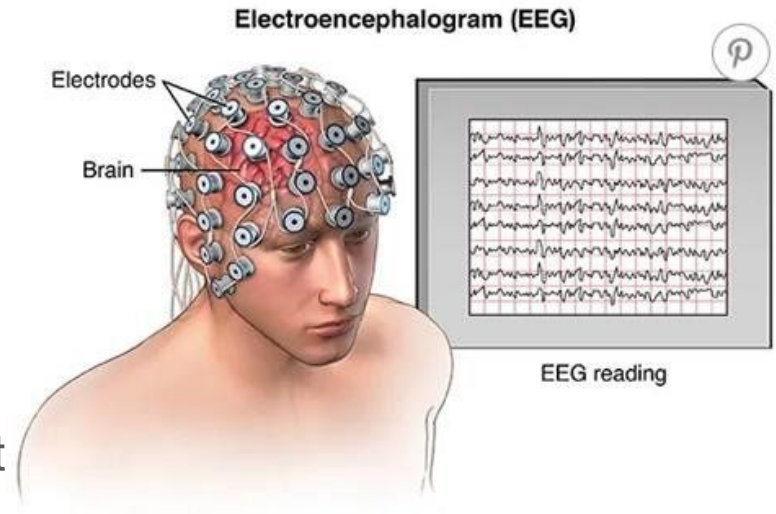
- By capturing a signal from multiple sensors we get a lot more information
- 12-lead ECGs given spatial resolution on cardiac abnormalities
- Increases analytical complexity (e.g., handling inter-channel covariance + autocorrelation)



Schreibgeschwindigkeit 25mm/sec

Electroencephalogram (EEG): many channels

- Electrogram of macroscopic brain activity measured from scalp (or intracranially).
- Signal from sets of neuron action potentials (ion-gated membrane de/repolarisation)
- Different electrode layouts/types impact signal resolution



Electroencephalogram (EEG): defined frequency bands

- EEG signals get divided into defined frequency bands
- Different brain activity typified by band of majority of activity (e.g., Delta -> Deep Sleep).
- Patterns in EEG can diagnose neurological disorders (including sleep disorders and epilepsy)

Beta
[12-30 Hz]



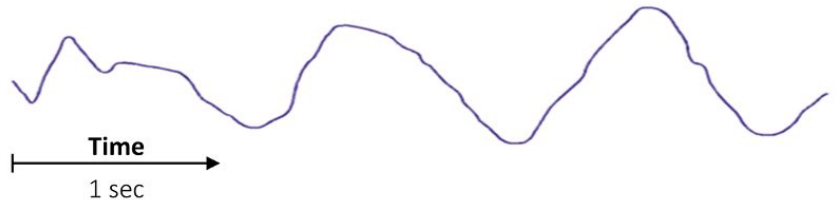
Alpha
[8-12 Hz]



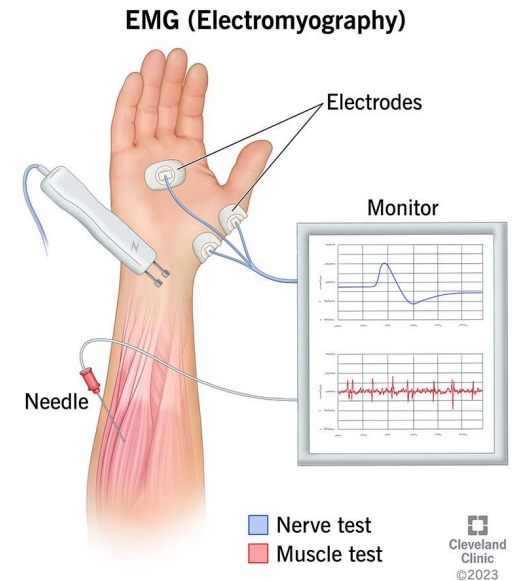
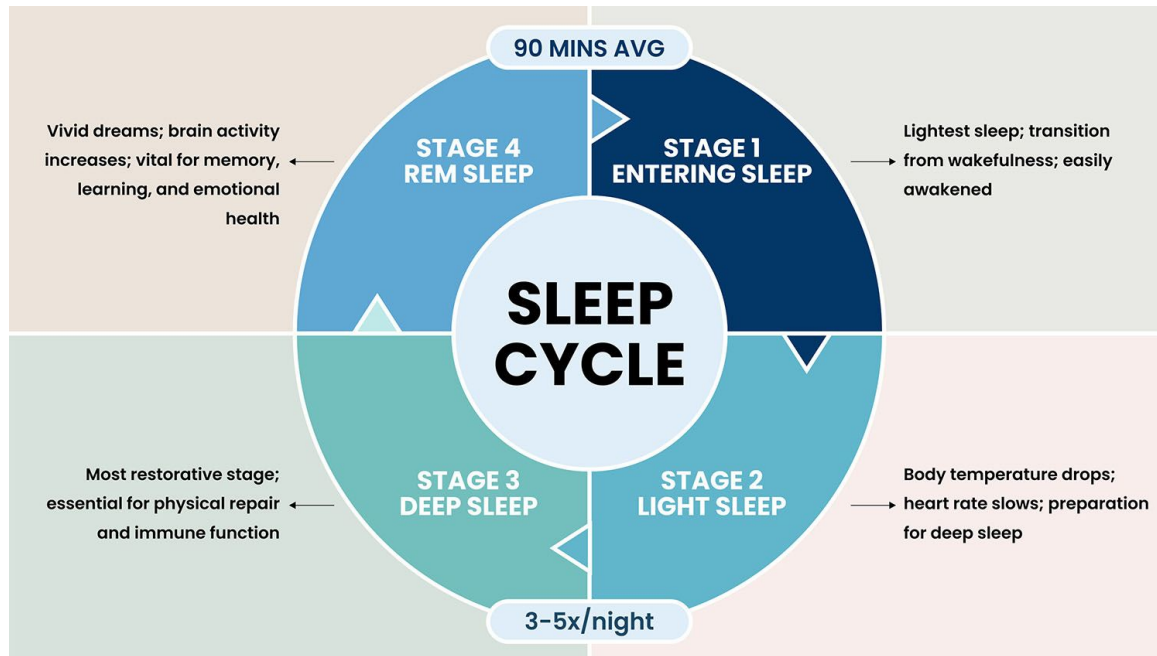
Theta
[4-8 Hz]



Delta
[1-4 Hz]



Polysomnography: EEG + Eye Tracking + Muscles



- Sleep split into defined stages (N1, N2, N3, REM) with distinctive EEG, eye movement, and muscle patterns.
 - **N1:** alpha + theta, slow eye movement
 - **N2:** theta + beta spikes, some movement
 - **N3:** delta, high amplitude, least movement
 - **REM:** sawtooth complex waves, rapid eye movement
- Sleep studies collect and analyse this data
- Predictive models can automate staging effectively (comparable IRR to human)
- Wearable EEG and movement tracking provide approximate to moderate sleep tracking at home



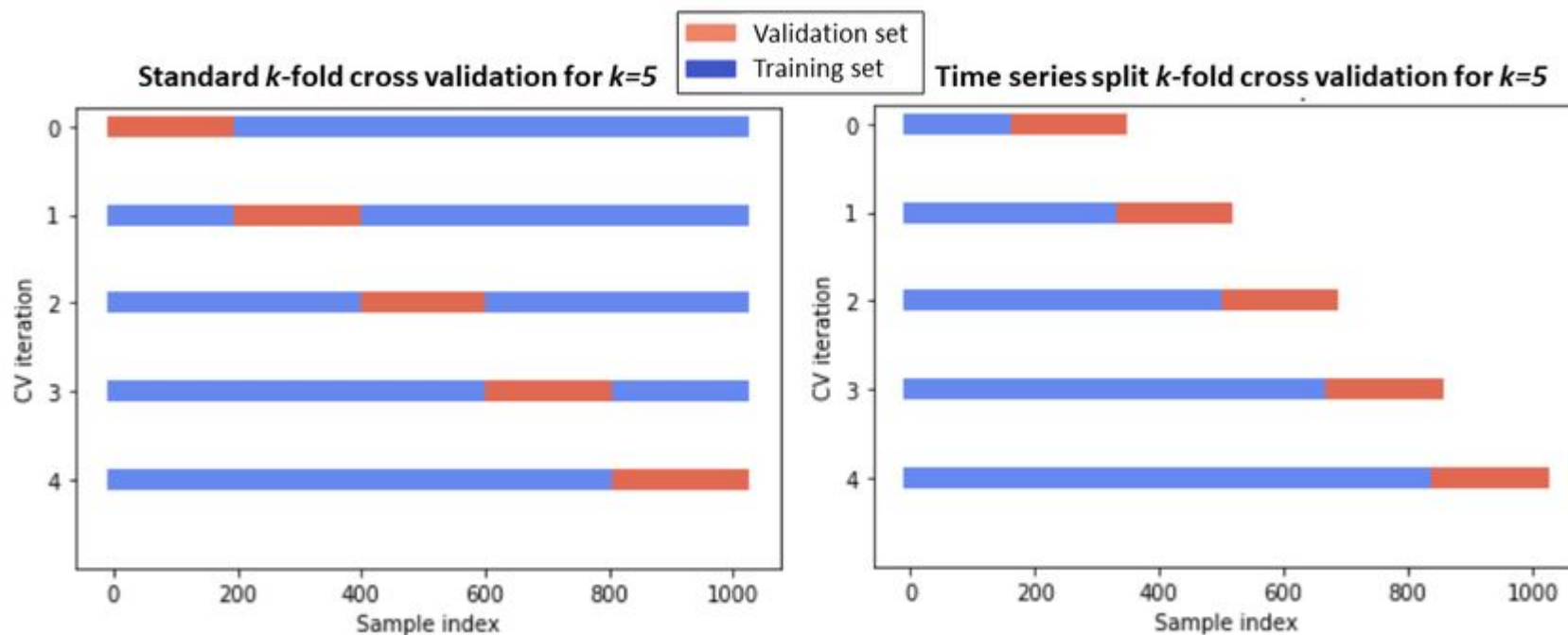
Why is analysing sensor data hard?

Challenges analysing physiological sensor data

- **Signal Quality**
 - Typically very noisy data susceptible to contamination
- **Data Volume and Sampling Frequency**
 - 32-lead EEG at 1000 Hz sampling rate creates 2.76 billion data points per day
- **Inter-individual Variability**
 - Normal resting heart rates ranges from 40-100 BPM
- **Temporal Dynamics**
 - Complex temporal patterns that change over multiple time scales
- **Context Dependence and State Transitions**
 - Meaning depends on context 150 BPM normal during exercise but pathological during rest

Statistical/ML Challenges of sensor data

- Statistical properties of data often change over time (non-stationarity)
 - core assumption in many models
- Data is not independent and identically distributed
 - core assumption in many models
- Easy to leak data during training and cross-validation

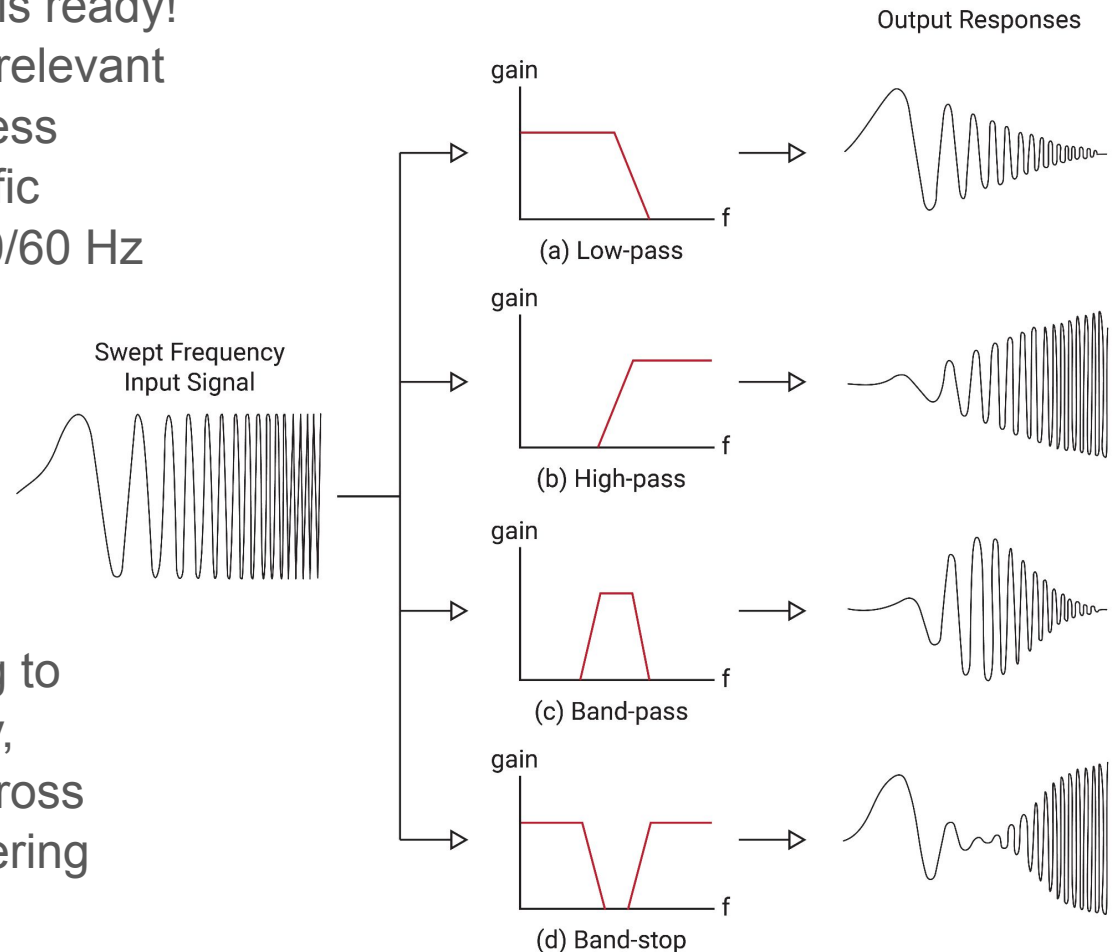


How do we actually analyse sensor data?

First, preprocessing!

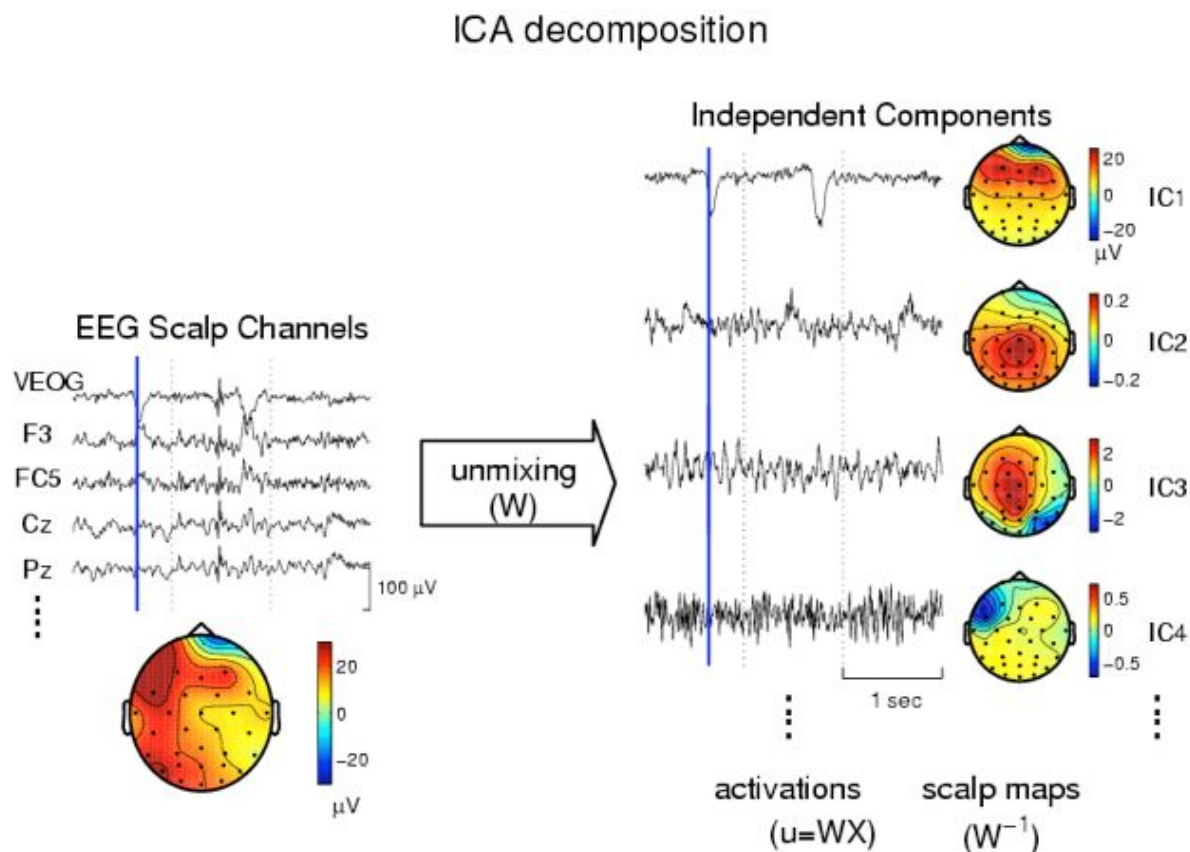
Removing noise/amplifying target signals

- Raw signals are not analysis ready!
- **Bandpass filter**: keep the relevant frequencies and drop the rest
- **Notch filter**: remove specific interference (common is 50/60 Hz from power lines).
- **Normalisation**: resampling to common capture frequency, amplitude normalisation across channel/person, artifact filtering



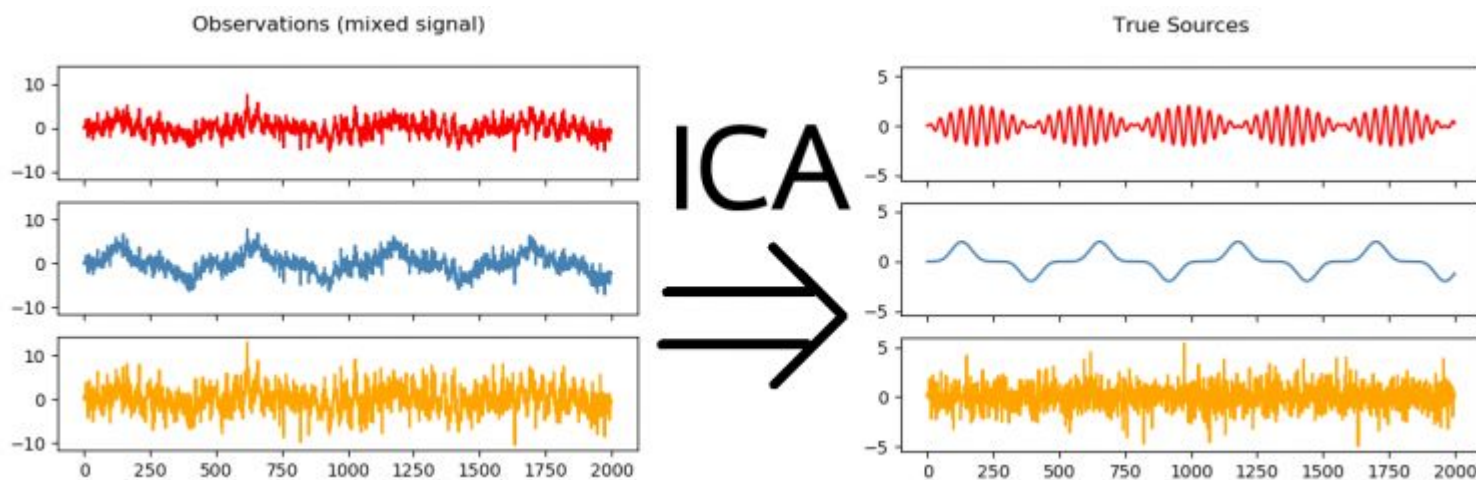
Decomposing data from multiple sensors

- Measured medical phenomena are often a mixture of signals from different sources
- Multiple sensors = each captures those sources (or a subset)
- Same source through different sensors will have different characteristics (amplitude, lag) due to sensor location



Decomposing data from multiple sensors: Independent Component Analysis

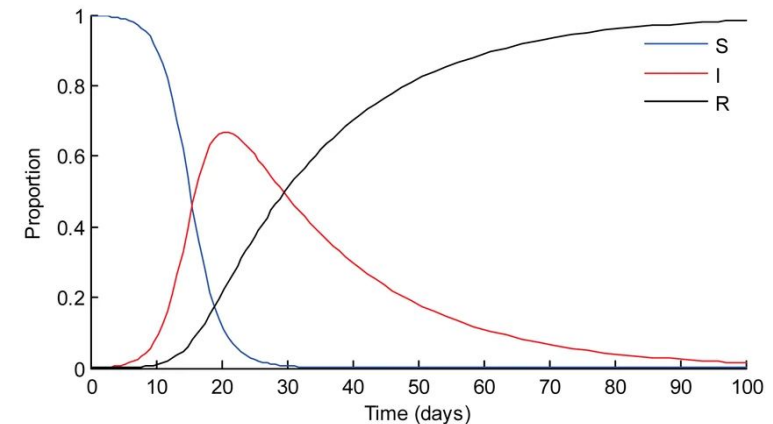
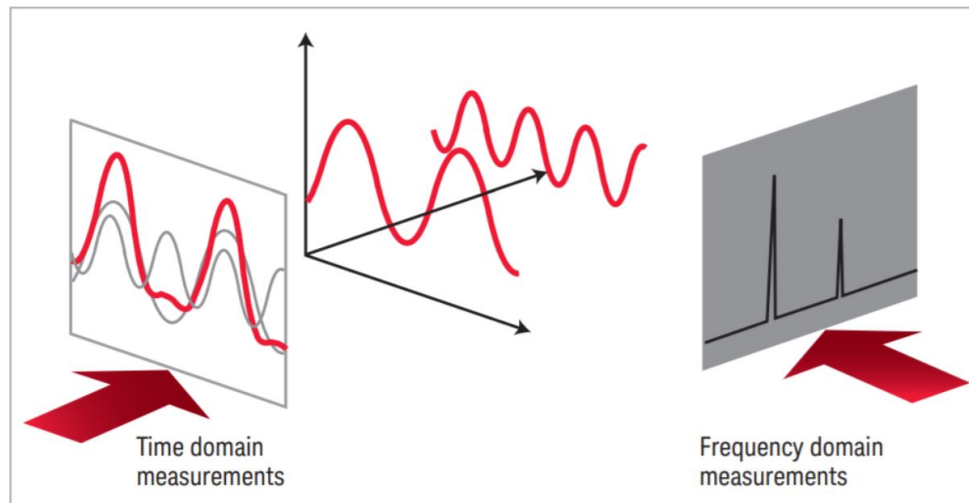
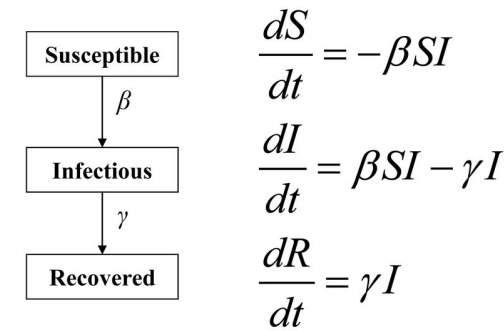
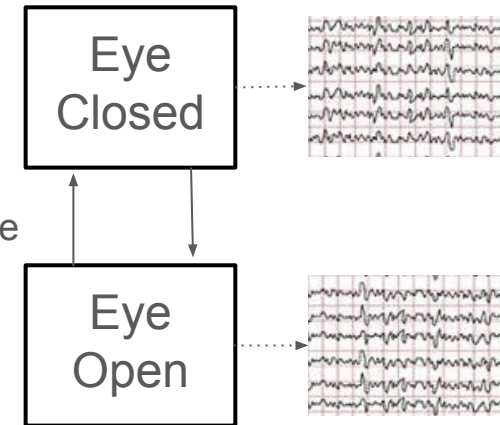
- Decompose signal into linear mixture of independent sources
- Part of most EEG analysis/processing workflows
- “Sphering” data (remove correlations between channels: cholesky decomposition with covariation matrix)
- Identify gaussian components of sphered (aka “whitened”) matrix



Now what?

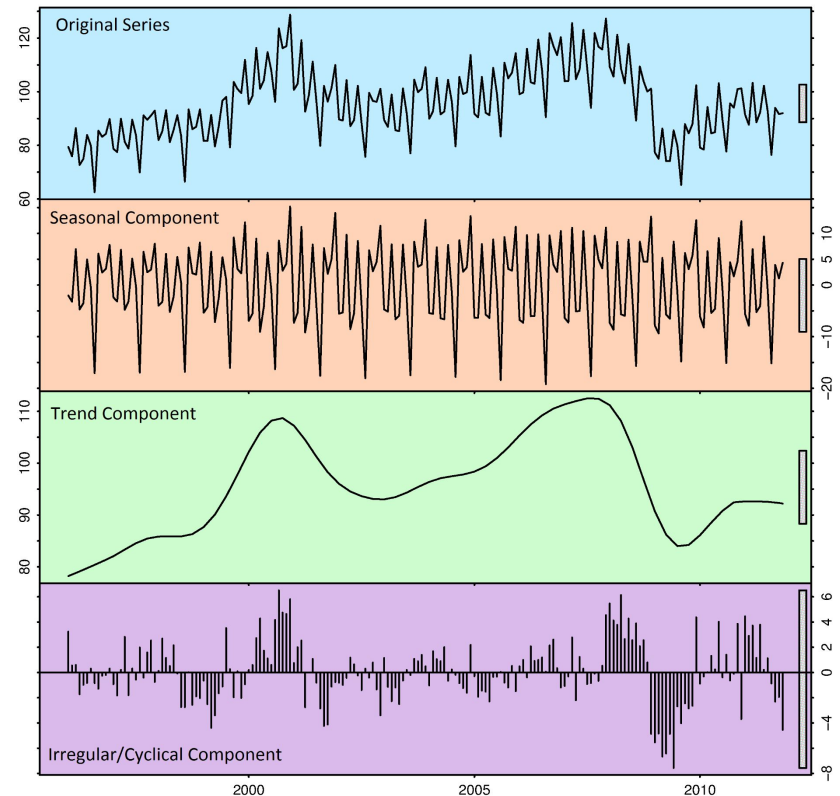
Analysis approaches for sensor data

- Moment (time domain) representation
 - Considering the statistical properties of the input data jointly over time
- Spectral (frequency domain) representation
 - Analysing the frequency-space representation
- Path (state space) representation
 - Describe the system as a dynamic system over time
- Change representation: systems of differential equations
 - Not going to discuss these but very common classical statistic or applied maths approach to sensor data.
 - Stochastic (SDEs) or deterministic (ODEs)



Time Domain: Decomposition

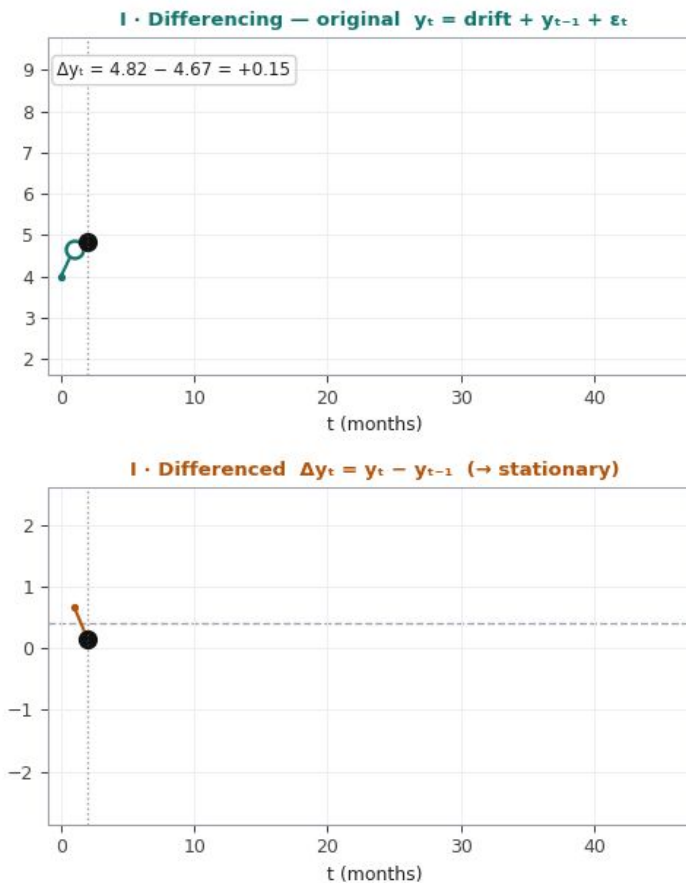
- Decomposition enables measuring strength of trend and seasonality
- Can be additive or multiplicative (are components independent?)
- Estimate trend/cycle using moving average
- **Moving average:** smooth series using mean/median over window (size = order)
- **Detrend series:** signal - moving average
- Moving average of detrended data: **seasonality**
- Multiplicative decomposition (divide rather than subtract)
- More advanced modern decomposition methods (EMD/SEATS/X11)



Time Domain: Differencing to remove trends

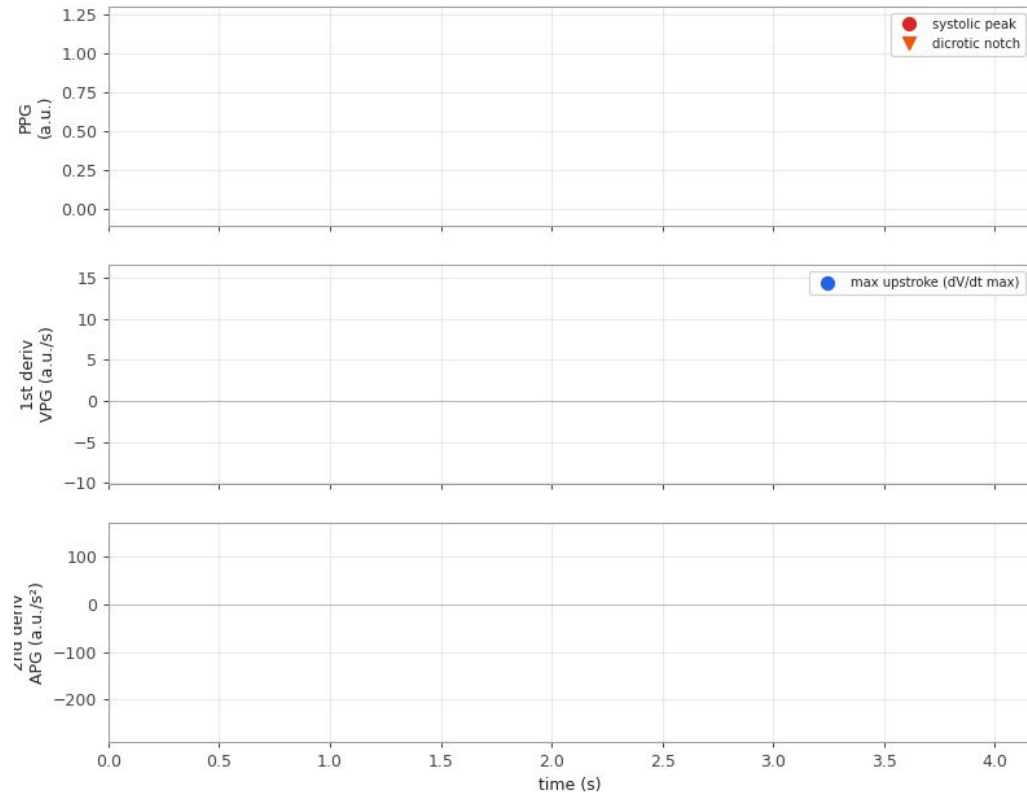
- **Differencing:** computed differences between consecutive observations:
 $\nabla X(t) = X(t) - X(t-1)$, $\nabla^2 X(t) = X(t) - 2X(t-1) + X(t-2)$

Degree of differencing
removes linear ->
higher order trends



Time Domain: Derivatives can also amplify signal changes

- **PPG:**
 - Systolic peak = peak blood volume
 - Dicrotic notch = aortic-valve closure
- Features hard to see in original **PPG**:
 - Differentiation acts as a feature amplifier.
- **1st derivative / velocity (VPG):** peak marks max upstroke
 - Informative of pulse-transit-time
- **2nd derivative / acceleration (APG):** b/a ratio
 - Correlates with arterial stiffness / vascular aging

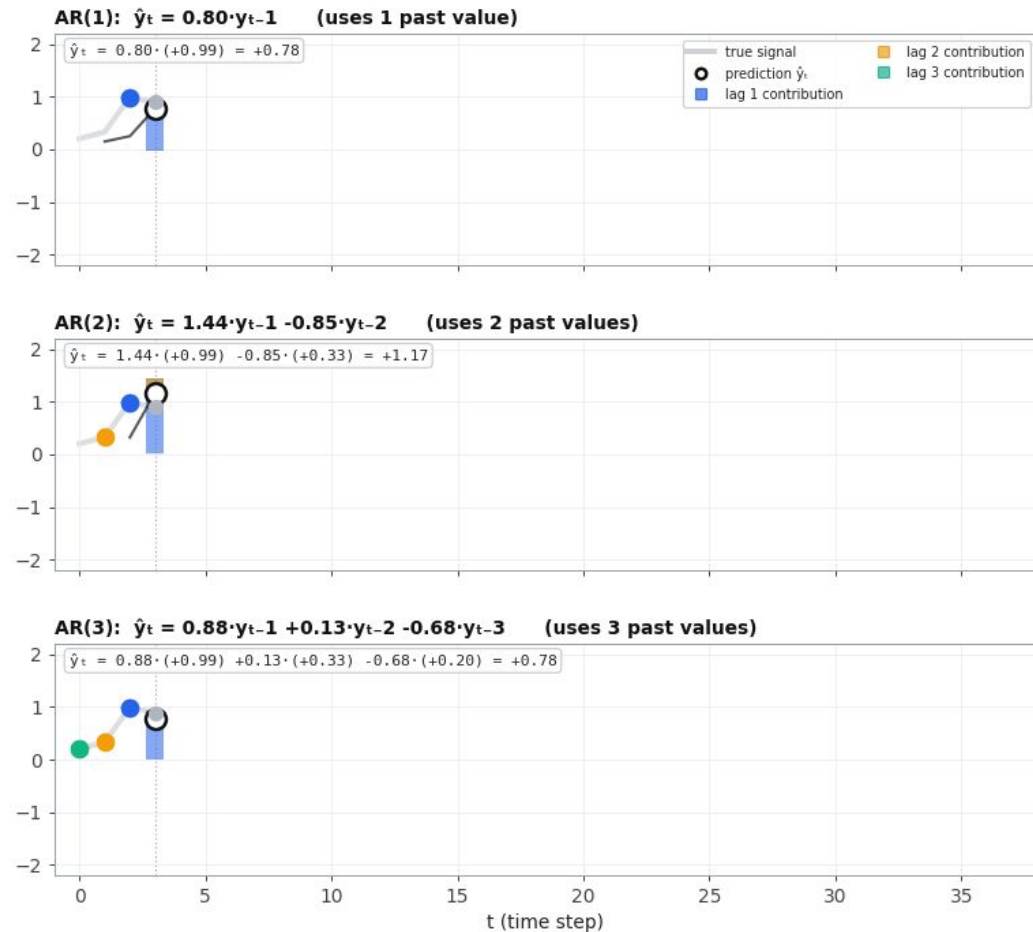


Time Domain: AutoRegressive models

- **AutoRegression:** Predict value at time t based on linear combination of past values of variable:

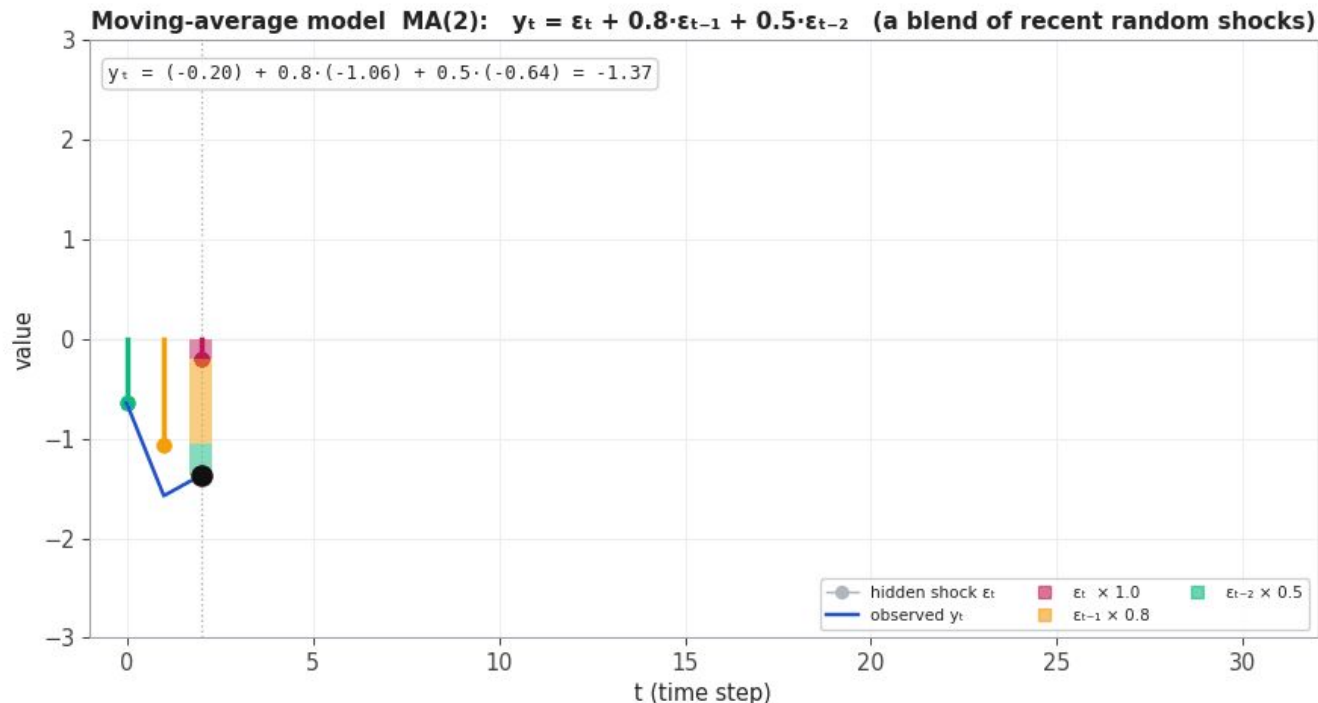
$$X(t) = \varphi_1 X(t-1) + \varphi_2 X(t-2) + \dots + \varphi_p X(t-p) + \varepsilon(t)$$

- Order of model is number of lagged values used
- $\varphi_1 = 0$ represents white noise
- $\varphi_1 = 1$ represents a random walk (with or without constant drift)



Time Domain: Moving Average models

- Instead of past values predict using past errors:
$$X(t) = \mu + \varepsilon(t) + \theta_1\varepsilon(t-1) + \theta_2\varepsilon(t-2) + \dots + \theta_q\varepsilon(t-q)$$
- For stationary data $AR(p) = MA(\infty)$
- Not to be confused with moving average smoothing used in decomposition



ARIMA: Combining Differencing, AR and MA models

- **ARIMA**(p,d,q) = AutoRegressive(p) + Integrated(d) + Moving Average(q)
- ARIMA combines differencing to achieve stationarity with ARMA modeling of the stationary series:
 - Difference d times: $\nabla^d X(t)$
 - Model stationary series: ARMA(p,q)
 - Integrate back to original scale
- Requires MLE / Information Criterion to fit orders
- Core of gold-standard time-series regression/forecasting method
- More advanced methods:
 - Vector Autoregression (VAR): enables feedback between forecasted variable and predictors (more realistic for real-world data)
 - Feed lagged values (or error) into ML model e.g., neural network with or without convolutions

$$y'_t = c + \underbrace{\varphi_1 y'_{t-1} + \dots + \varphi_p y'_{t-p}}_{\text{lagged values}} + \underbrace{\theta_1 \varepsilon_{t-1} + \dots + \theta_q \varepsilon_{t-q} + \varepsilon_t}_{\text{lagged errors}}$$

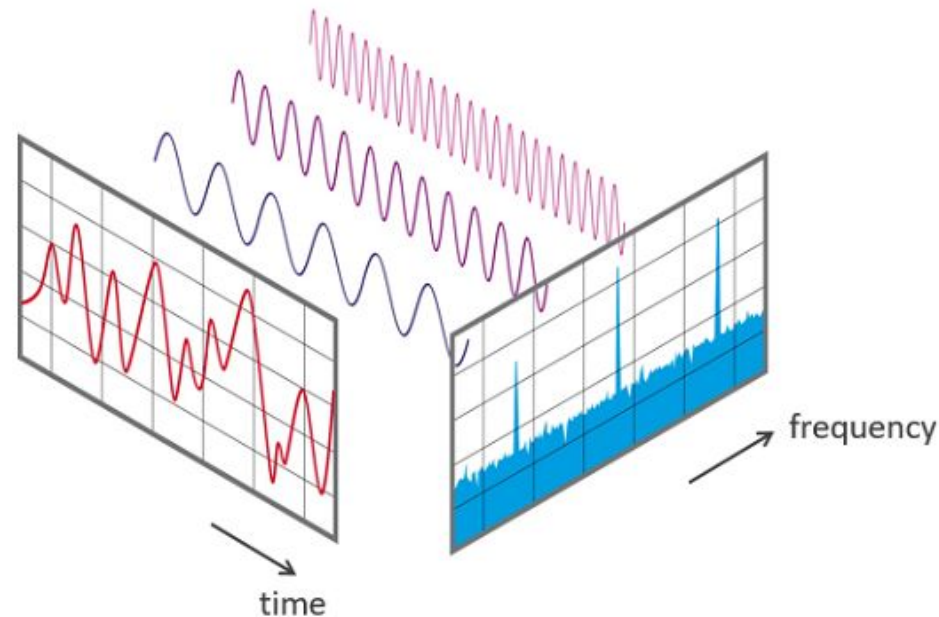
Diagram labels:

- intercept (points to c)
- lagged values (points to the AR part)
- lagged errors (points to the MA part)
- differenced time series (points to y'_t)

What about aggregating over frequency
instead of time?

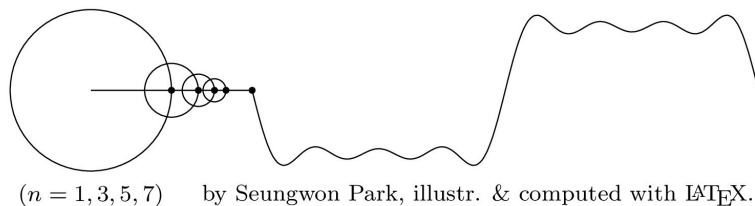
Frequency/Spectral Domain

- Signal composed of multiple frequencies (e.g., EEG power bands)
- Can greatly simplify analysis (offers simple decomposition)
- Feeds into many useful mathematical tools (resonance, harmonics, power spectral densities, eigenvalues, ...)
- Several different ways of converting time-domain to frequency-domain
- Laplace and Fourier methods are most common

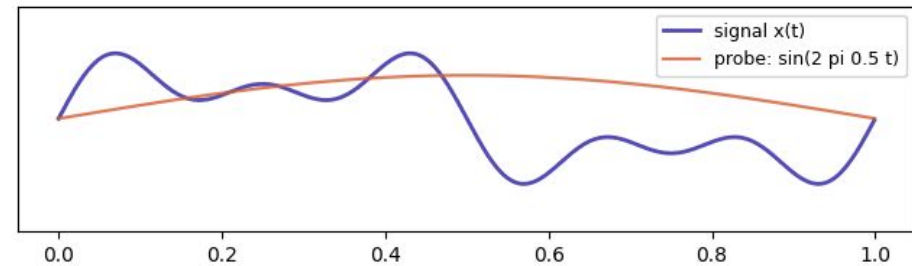


Frequency Domain: Fourier Transform

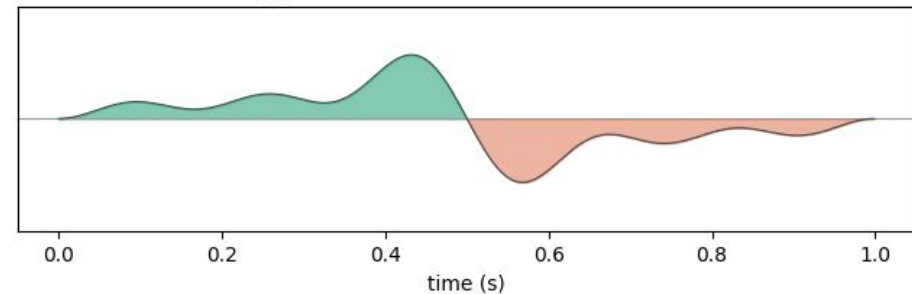
- Fourier Transform:
Time -> Frequency
- Inverse Fourier Transform:
Frequency -> Time
- Decompose signal into series of angular components
$$x(t) = a \sin(2\pi f t + \varphi)$$
- Location (frequency) and height (amplitude) of frequency spectra peaks can be used as input for whatever model you want.



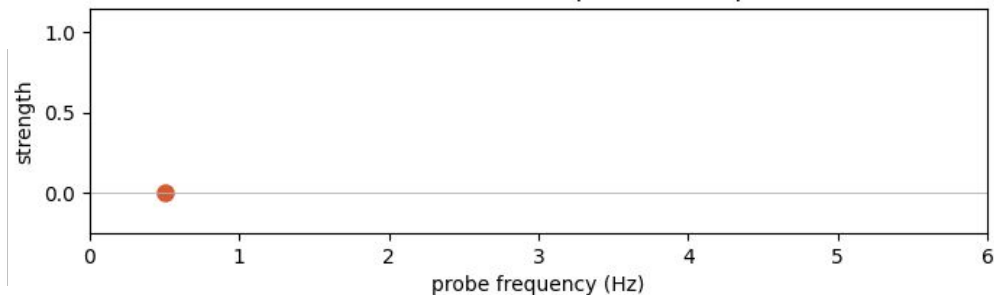
1. Lay a probe wave over the signal



2. Multiply, then sum the area -> value = +0.00

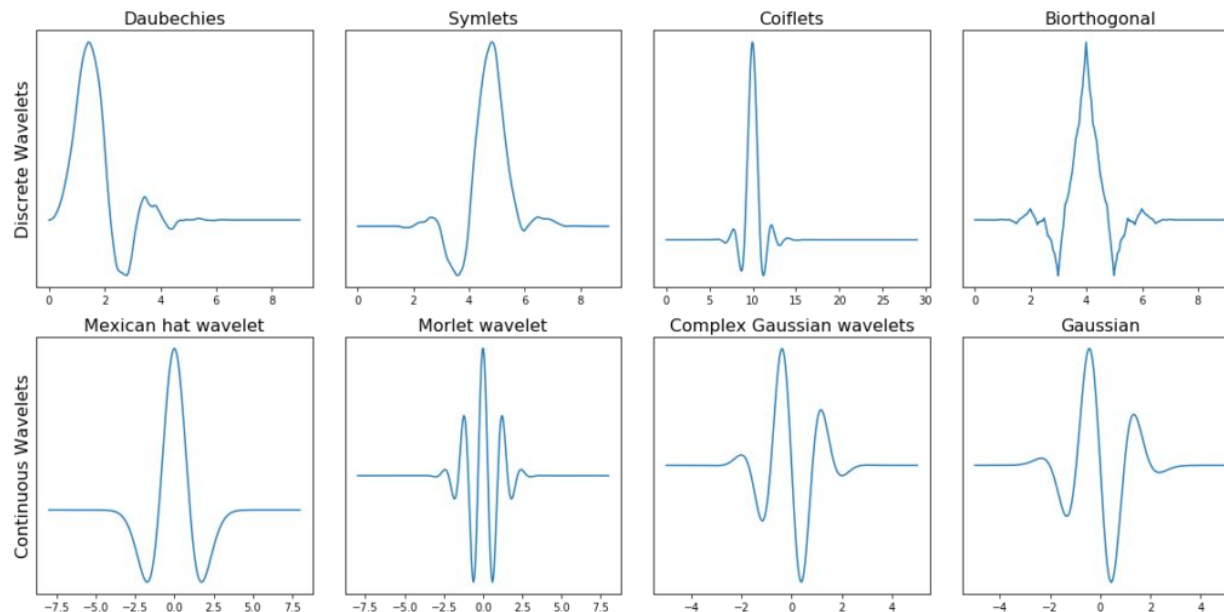


3. Each area becomes one point of the spectrum



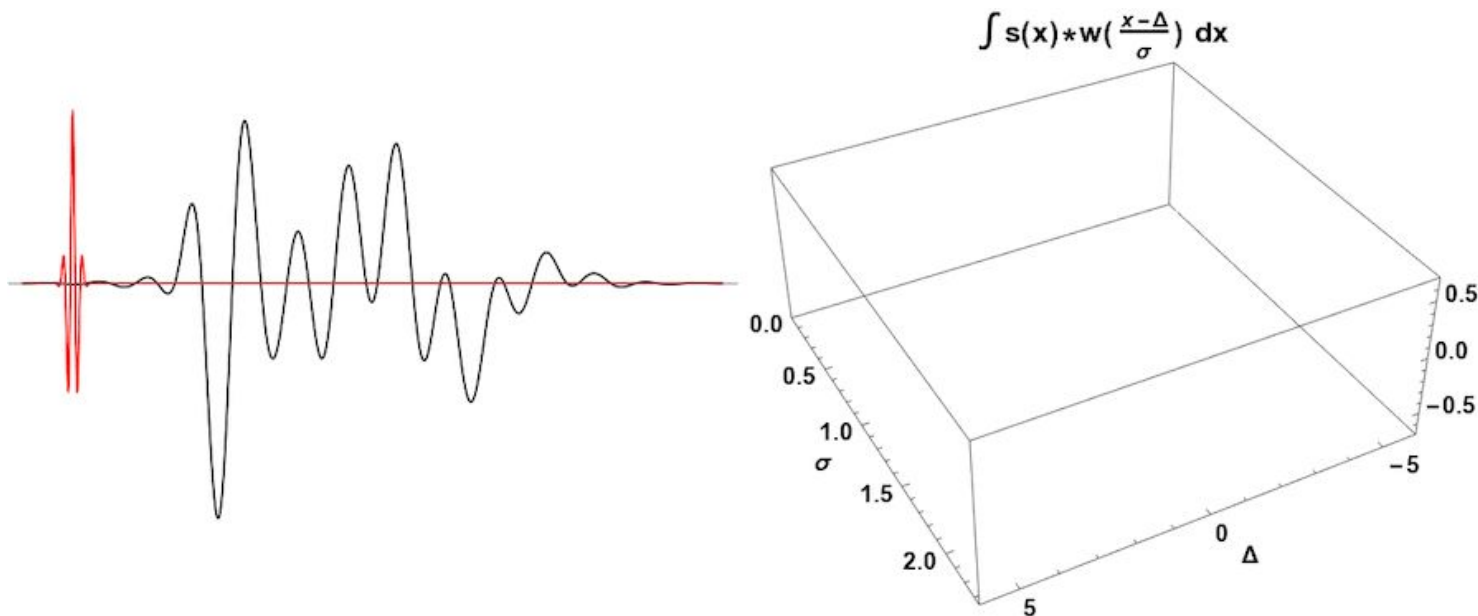
Time-Frequency Domain: Wavelet Transforms

- Fourier transform has great frequency resolution but no time resolution
- Wavelet allows retaining frequency and time resolution: capture dynamic frequency spectra within signal
- Convolve signal with variety of waves (wavelets) with scale (frequency) and location (time) properties
- Wavelet can be learnt in the same way as convolutional kernels



Time-Frequency Domain: Wavelet Transforms

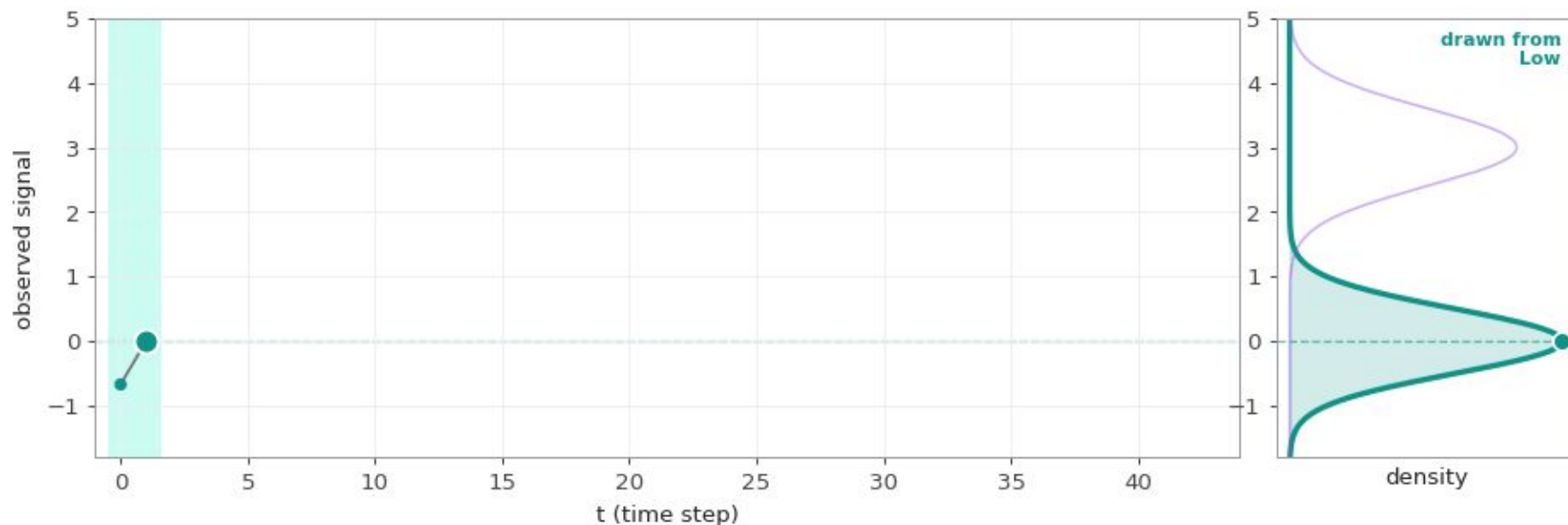
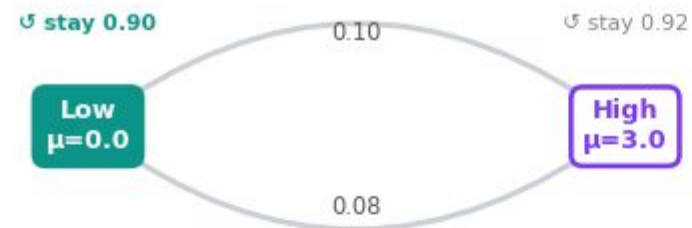
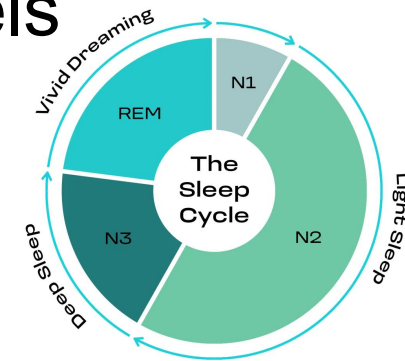
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Signals (ideally) reflect an underlying state so
why not just model those states directly?

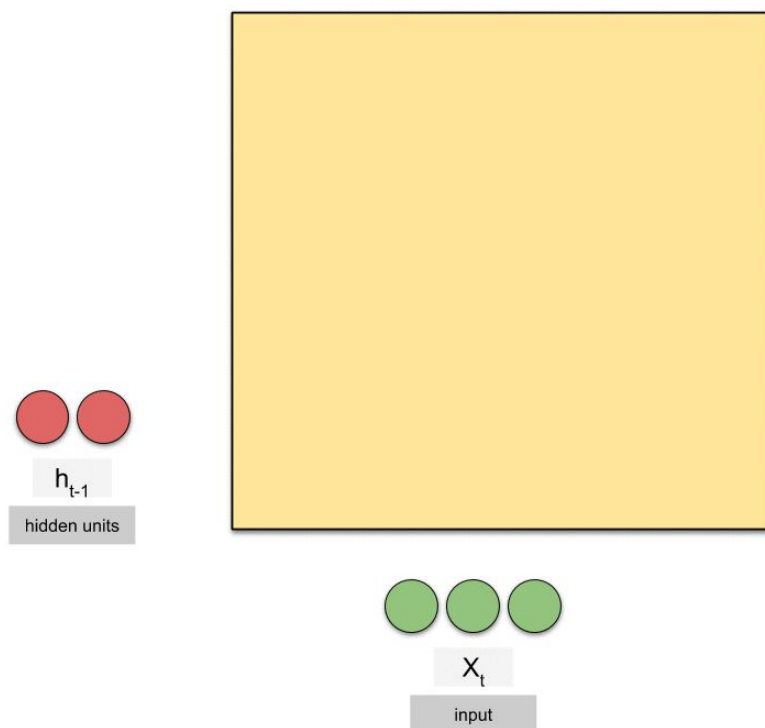
State-space models: Hidden Markov Models

- Data is represented resulting from a series of hidden states
- Model describes movements between hidden states
- Observed values are derived from hidden states
- Markov property: only previous state(s) matter
- More naturally discrete time
- Continuous time is possible though
- Well suited to classification/detection

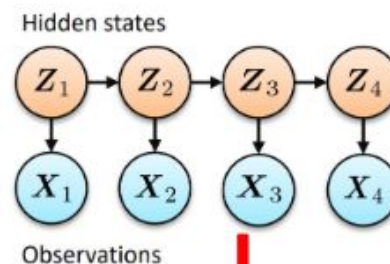


Going beyond HMMs: RNNs and Transformers

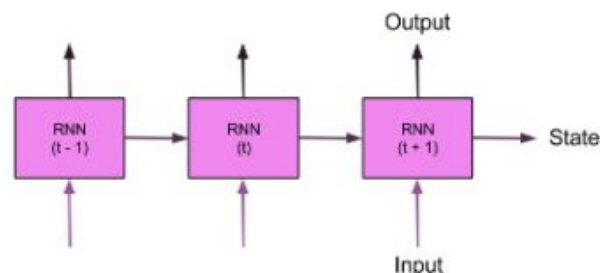
- Recurrent Neural Networks and Transformers can act like HMMs with more complex dependencies
- Alternative state space models: best of both worlds
- Attention mechanism similar to soft/variable-order HMMs



Maintain probabilistic structure of HMMs



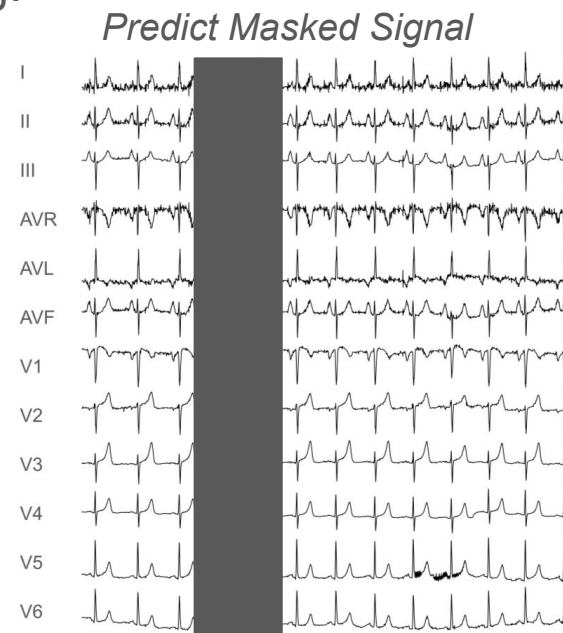
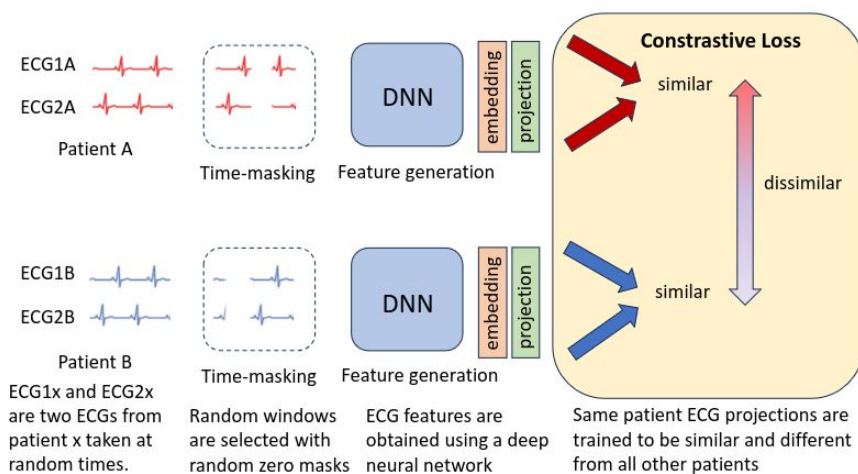
But use RNNs to model state dynamics



- State-space models (e.g., Mamba/S4) are the current state of the art in this space.

Generalised Representations: Signals Foundation Models

- But, good physiological signal data is rare so how do we get learn good representations for deep learning?
 - Data itself is noisy with high variability across people, devices, capture modalities.
 - Labelling is challenging (need expertise, moderate to high inter-rater reliability)
- Solution: use massive (>5 million series) datasets with augmentation & diffusion to try to learn generalisable features in self-supervised context
- Multiple samples per person for contrastive learning.
- BrainBERT (EEG), TA-PCLR (ECG)
- **BUT:** scaling not as clean as text or image models



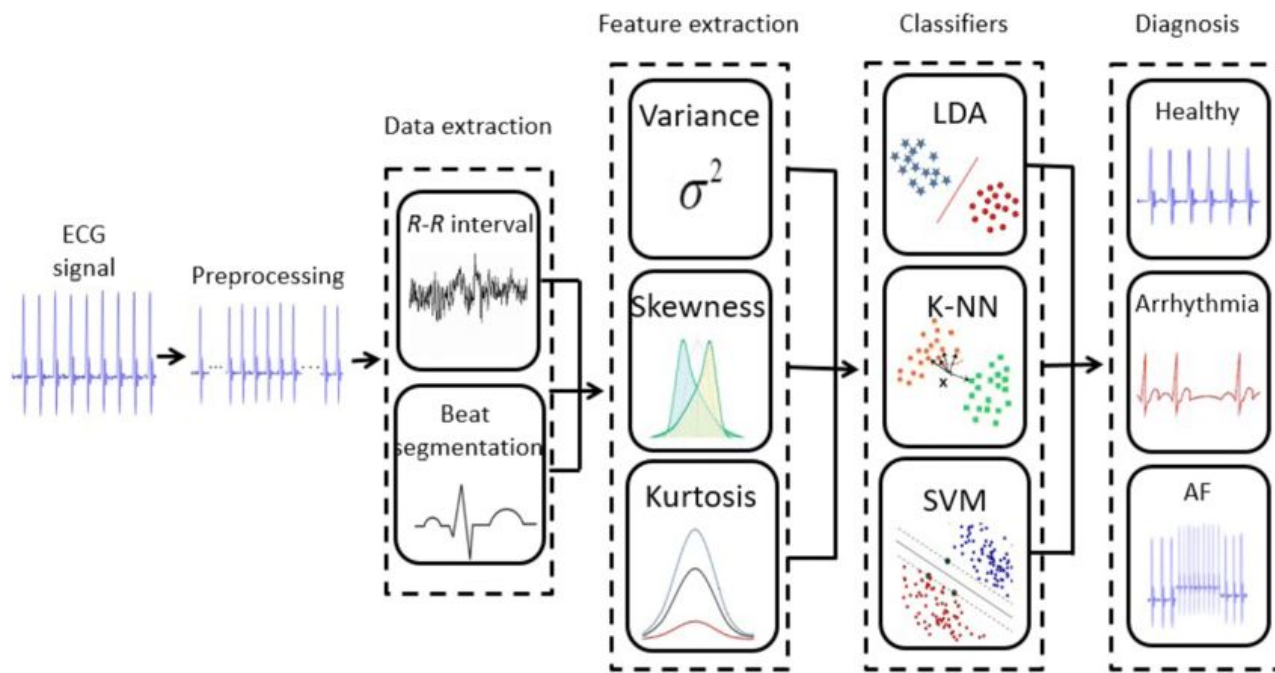
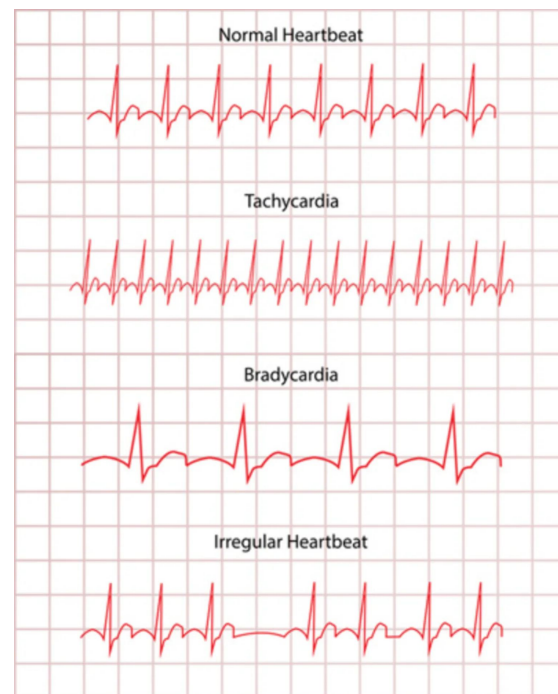
(a) 12-lead ECG

So, how are these methods actually being used?

ECG arrhythmia classification

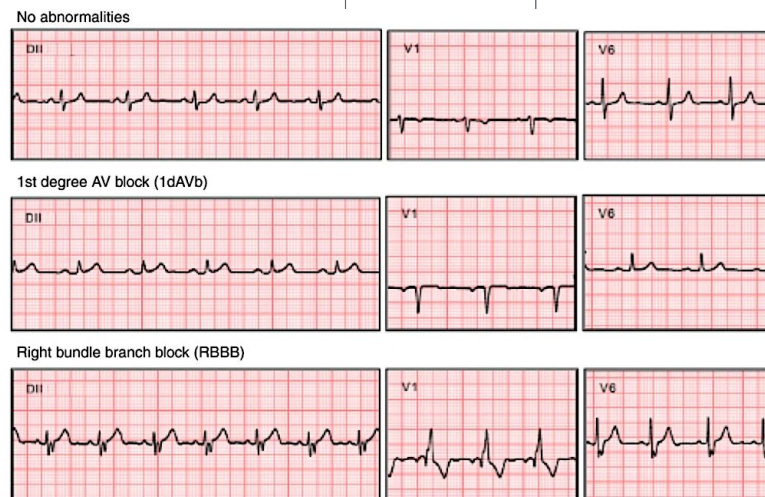
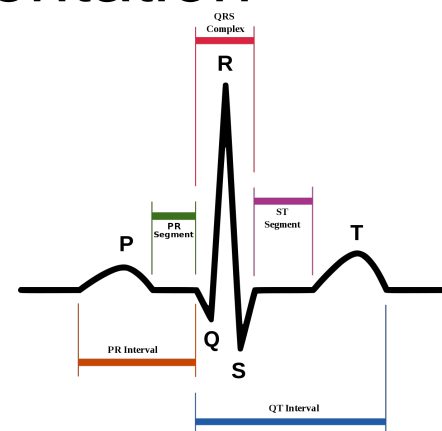
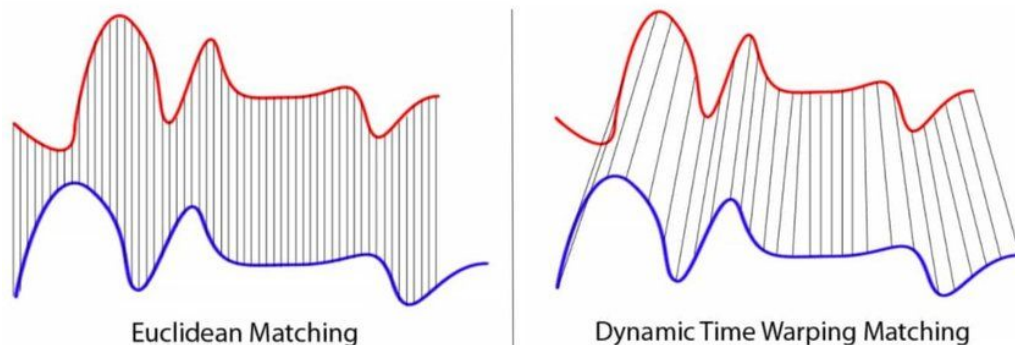
MIT-BIH Arrhythmia Database:

- Classic Benchmark
- 48 x 30-minute 2-lead 360 Hz ECGs
beat-level annotations
- Denoising -> Segmentation -> Model
- Normal Beats >>> Abnormal Beats



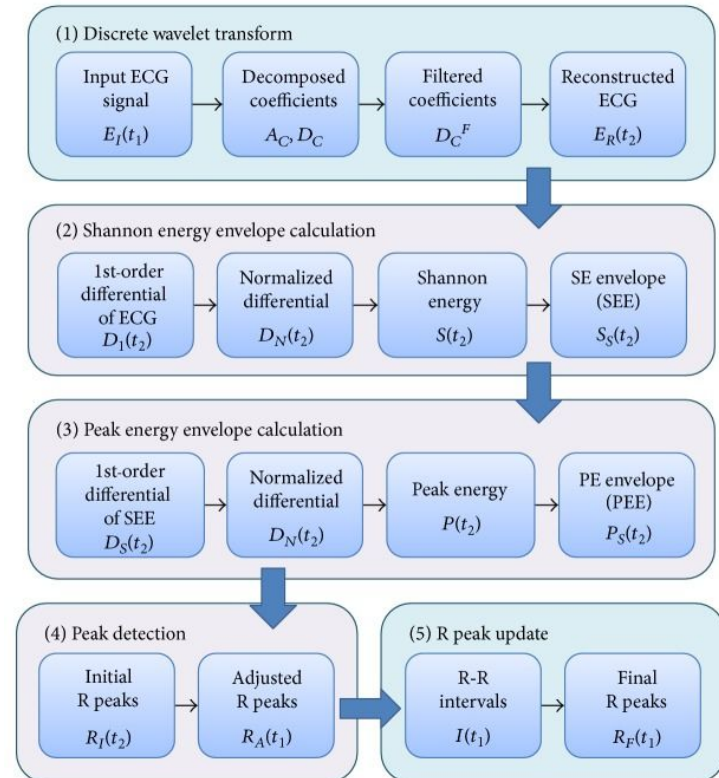
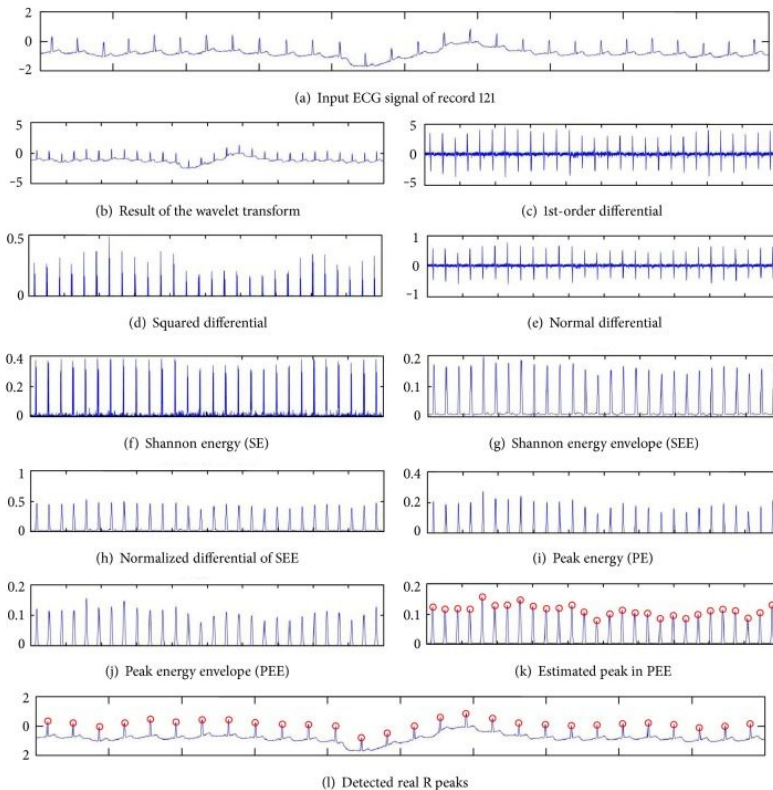
Dynamic Time-Warping beat segmentation

- Clean ECG: just identify highest peaks but ECG is often noisy
- Know what a heartbeat looks like: align to ECG
- Often detecting arrhythmias/abnormal heartbeats: may not align
- Allow time to be “fuzzy” in alignment: dynamic time warping



Wavelet Transform Beat Segmentation

- Wavelet transforms can make R-peaks very clear even in noisy data
- Hand-crafted features can then be extracted
- Alternatively deep models can be used to learn wavelets and segmentations (unsupervised or supervised)



Electrocardiogram (ECG)

Many ECGs out there ranging from “deployed” to

Not really clinically useful;
Only needed at extremes/odd
situations (administrative
cut-offs) where error is highest

- Predict age and sex
- Detect anaemia (>90% accuracy with demographic data)
- Predict likelihood of low ejection fraction
- Automated detection of amyloid heart/cardiomyopathy/mitral valve prolapse
- Predict 1-year mortality (AUC > 0.85)

“with demographic data” - is
ECG contributing much?
Improvement over CBC?

What makes a prediction useful?

Actionable?

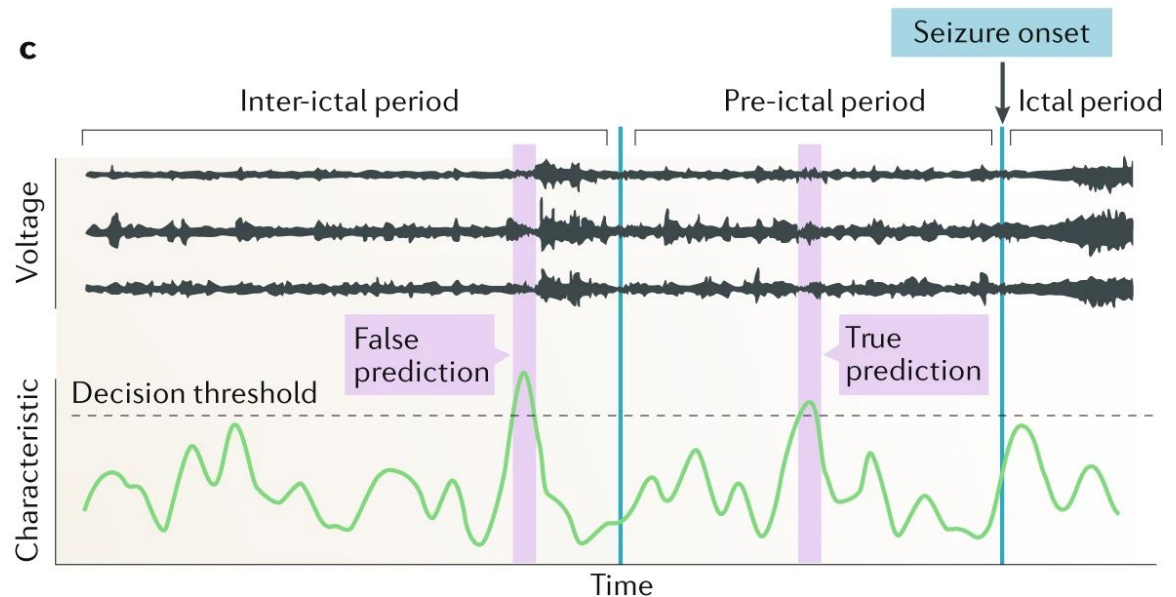
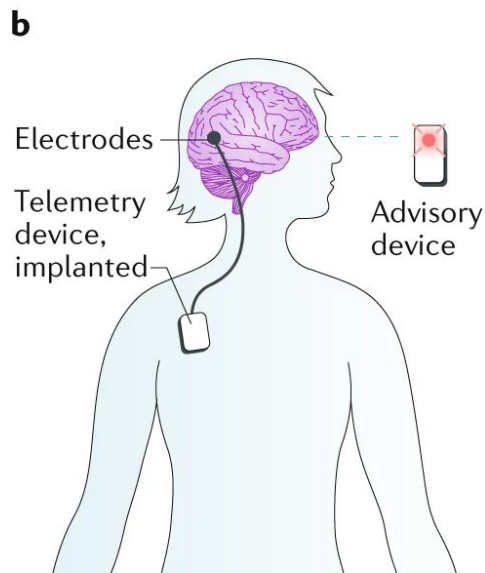
- Does a prediction actually guide an action?
- Is it replacing a test that is already good enough?
- Is there a useful PPV at deployed prevalences?

Rare at population level, so
very conditionally useful

Cluster-Randomised Trial
(EAGLE)
Earlier low-EF interventions

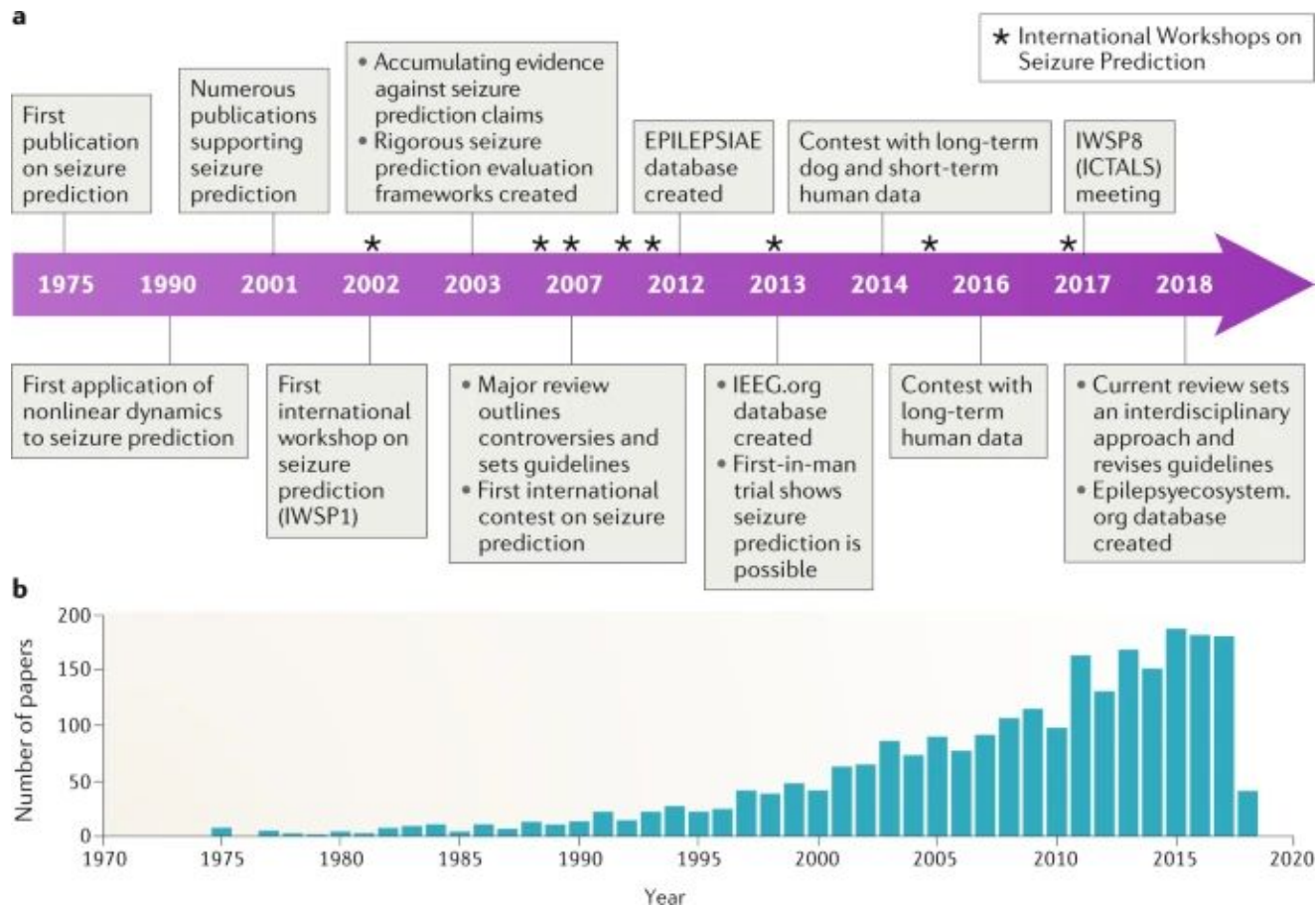
Predicting epileptic seizures

- Epilepsy has a global prevalence of 1% (80 million)
- 30% of cases not treatable with anti-epileptic medication (24 million)
- Unpredictability of seizures is major source of mortality and morbidity
- Permanent intracranial EEGs now possible



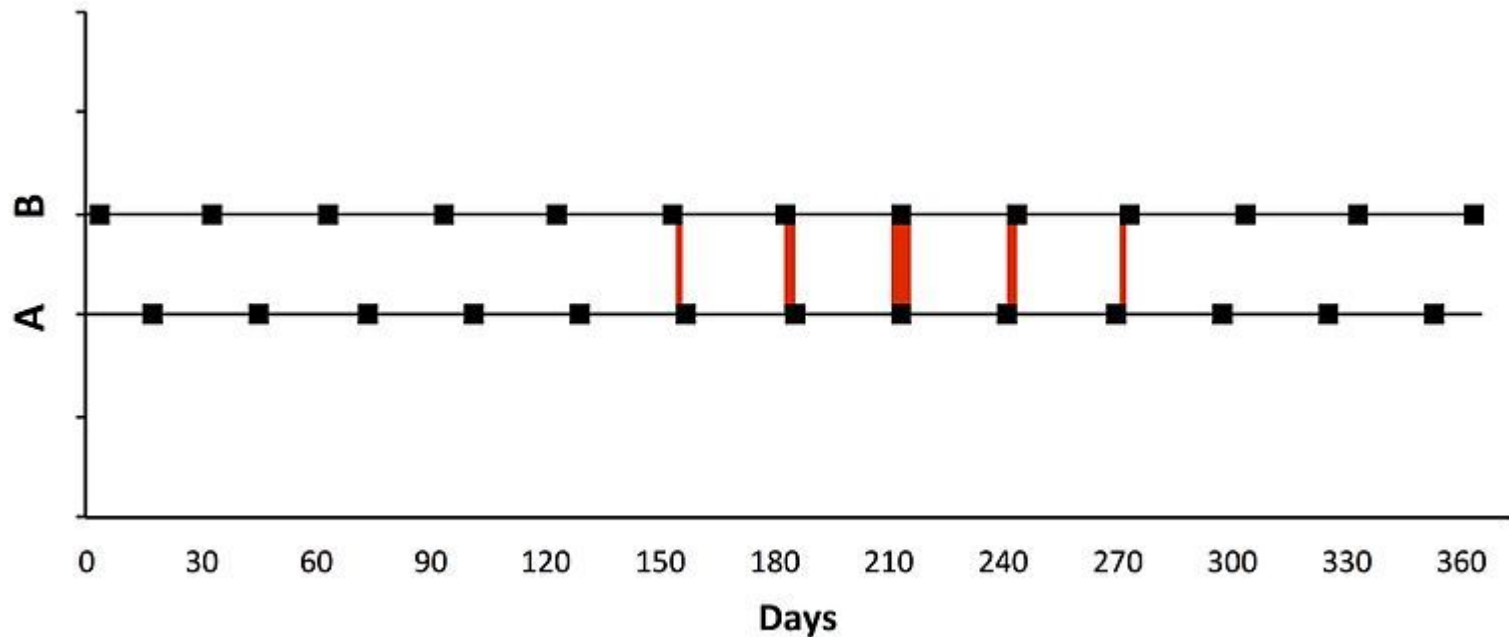
Lots of research

- 2007 review: insufficient evidence that seizures can be predicted



Nulls for periodic signal can be challenging

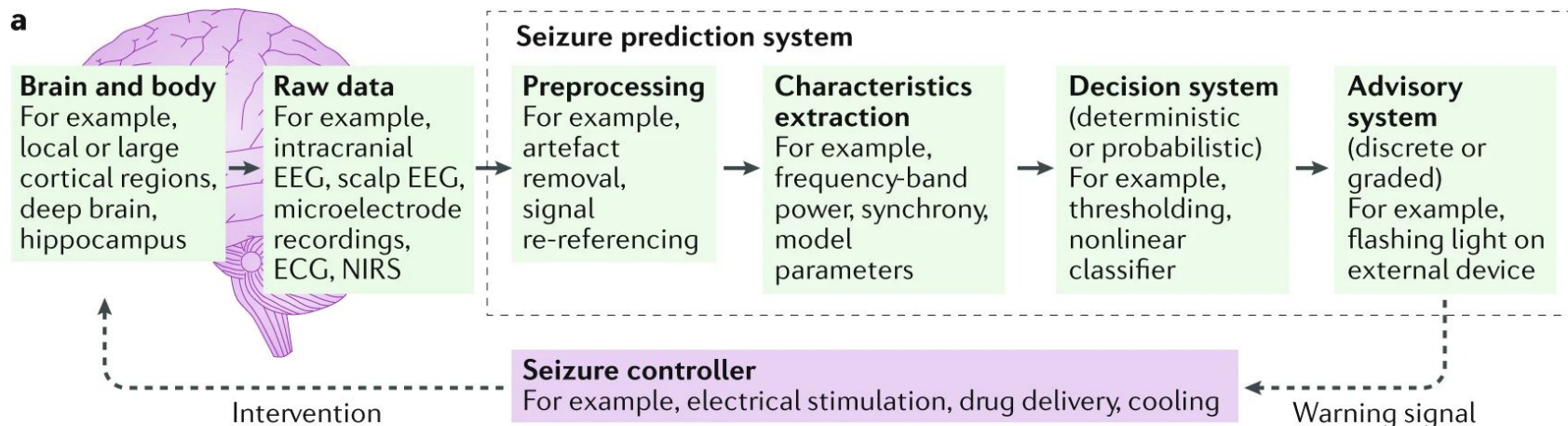
- Randomly shuffle seizure onset times => no pre-2007 actually worked



https://en.wikipedia.org/wiki/File:Yang_and_Schank_2006_converging_diverging_cycles2.jpg

Most take a similar approach

- Inherently unbalanced data (seizures are rare compared to interictal EEG)
- Non-continuous datasets (i.e., inter-ictal and pre-ictal chunks) can make task easier than reality
- Scoring predictions is challenging (prediction window / time to seizure onset)
- Suitable baseline performance metric (random prediction: diurnal?)
- Inter-person variance can be large (electrode placement, cycles)



American Epilepsy Society Seizure Prediction Challenge

- Winning entry: well-crafted features for GLMs (averages with RF)
- General approach: bunch of features into large ensemble models
- 2014 (still relatively early days for CNNs being used outside of images)

	Dog_1	Dog_2	Dog_3	Dog_4	Dog_5	Patien	Patien	all
SVC_ica_psd_logfBB_AND_ica_xcorr-tpeak	0.8175	0.9253	0.8129	0.7371	0.8617	0.8346	0.7526	0.8493
SVC_ica_ilingam-causalindex_AND_ica_PSDlogfcorrcoef	0.8125	0.9904	0.7356	0.6645	0.8540	0.9355	0.4759	0.8362
SVC_ica_PSDlogfcorrcoef_AND_ica_pwling5	0.8101	0.8916	0.7657	0.6663	0.9073	0.8236	0.5182	0.8280
SVC_ica_cov_AND_ica_lmom-3	0.8081	0.9383	0.8109	0.6689	0.9018	0.7532	0.6210	0.8458
SVC_ica_corrcoefeig_AND_ica_PSDlogfcorrcoef	0.8071	0.9858	0.7791	0.6792	0.8970	0.9208	0.4920	0.8456
SVC_ica_psd_logfBB_AND_ica_PSDlogfcorrcoef	0.8063	0.9856	0.8033	0.7442	0.8477	0.9002	0.7618	0.8618
SVC_ica_ilingam-causalorder_AND_ica_psd_logfBB	0.8059	0.9685	0.8166	0.7623	0.8398	0.8660	0.8117	0.8596
SVC_ica_ilingam-causalindex_AND_ica_psd_logfBB	0.8049	0.9783	0.8133	0.7610	0.8429	0.8653	0.8020	0.8619
SVC_ica_lmom-3_AND_ica_PSDlogfcorrcoef	0.8011	0.9816	0.7282	0.6717	0.8756	0.9547	0.4971	0.8400
SVC_ica_lmom-2_AND_ica_psd_logfBB	0.8008	0.9755	0.8358	0.7575	0.8591	0.8486	0.8169	0.8643
SVC_ica_ampcorrcoef-alpha-eig_AND_ica_pib_ratioBB	0.8004	0.9546	0.8612	0.7408	0.8666	0.8825	0.7204	0.8584
SVC_ica_pib_ratioBB_AND_ica_pwling5	0.7991	0.8737	0.8724	0.7281	0.8607	0.8163	0.5582	0.8353
SVC_ica_gcaus_AND_ica_pib_ratioBB	0.7973	0.9770	0.8548	0.7212	0.8296	0.8927	0.7014	0.8566
SVC_ica_lmom-4_AND_ica_psd_logfBB	0.7962	0.9711	0.8367	0.7613	0.8446	0.8588	0.8186	0.8613
SVC_ica_ampcorrcoef-high_gamma_AND_ica_phase-beta-sync	0.7921	0.9450	0.7474	0.6594	0.9699	0.9032	0.5327	0.8439
SVC_ica_ampcorrcoef-low_gamma_AND_ica_psd_logfBB	0.7899	0.9449	0.8367	0.7463	0.8714	0.8434	0.7935	0.8506
SVC_ica_ampcorrcoef-alpha-eig_AND_ica_phase-beta-sync	0.7889	0.9789	0.7460	0.7239	0.9605	0.9157	0.5430	0.8559
SVC_ica_ampcorrcoef-high_gamma-eig_AND_ica_corrcoef	0.7873	0.9320	0.8261	0.6221	0.9360	0.7966	0.6627	0.8563
SVC_ica_PSDlogfcorrcoef_AND_ica_xcorr-ypeak	0.7869	0.9826	0.7565	0.6327	0.9433	0.8982	0.7139	0.8641
SVC_ica_psd_logf_AND_ica_PSDlogfcorrcoef	0.7866	0.9816	0.8291	0.7369	0.8993	0.8998	0.7395	0.8725
SVC_ica_phase-beta-sync_AND_ica_pib	0.7770	0.9806	0.8304	0.6789	0.9556	0.9166	0.6674	0.8626
SVC_ica_ampcorrcoef-beta AND_ica_phase-beta-sync	0.7737	0.9440	0.7624	0.7355	0.9632	0.8947	0.5755	

High variance clinical trials: implementation science is key

- 3-100% accuracy across ≥ 3 seizures across individuals
- Seizures are non-random (short and long-term temporal dependence)
- Diving into why they don't haven't worked:
 - Individual seizure frequency
 - long-term temporal variations in seizure frequency
 - multimodal distributions of seizure duration and inter-ictal intervals

Lessons learnt:

- EEGs give poor mechanistic insight
- Emerging ideas about how seizures work: excitation/inhibition imbalance vs aberrant behaviour emerging from network parameters
- Implementation science is often more important than underlying ML

Learning Overview

- Types of medical sensor data: clinical (ECG, EEG) and wearable (PPG, accelerometry, SpO₂)
- Attributes of Signals: time, analogue/digital, sampling rate
- Challenges of Signals: noise, data volume, non-stationarity, non-IID, leakage
- Preprocessing: filtering, artifact removal, feature engineering (e.g. HRV)
- Multiple sensors: ICA decomposition
- Time-domain approaches: differencing, derivatives, AR / MA / ARIMA
- Frequency & time-frequency: Fourier and wavelet transforms
- State-space approaches: HMMs, RNNs, Transformers, Structured State-Space (S4/Mamba)
- Self-supervised learning and foundation models
- Examples:
 - Sound Cough/COVID detection (spurious ML)
 - ECG beat segmentation (dynamic time warping, wavelets)
 - EEG seizure prediction (nulls for signals/time-series, cyclical, implementation)